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Distributed Secure State Estimation Against Byzantine Agents: A Multi-hop Relay-based Sorting and Filtering Method

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***Abstract*—This paper addresses the problem of distributed secure state estimation (DSSE) for cyber-physical systems, where Byzantine agents are present in a multi-agent network. To achieve DSSE in polynomial time, a novel multi-hop weighted distributed gradient descent (MW-DGD) algorithm is proposed. Specifically, the algorithm integrates the sorting and filtering techniques with the minimum message coverage to construct a mathematical equivalent representation under multi-hop communication. Within this framework, we further derive a more general robustness condition. The resulting necessary and sufficient condition demonstrates that while multi-hop communication may expand adversarial attack surfaces, it ultimately enhances network resilience against Byzantine agents. Additionally, a more versatile distributed error detection algorithm is introduced to dynamically evaluate the estimation performance under Byzantine agents in multi-hop communication. Compared with existing methods, the MW-DGD algorithm eliminates the need for brute-force search, thereby significantly reducing computational complexity. Finally, simulation results validate the superiority of the algorithm.**

***Index Terms*—Cyber-physical systems; Distributed secure state estimation; Byzantine agents; Multi-hop communication; Distributed gradient descent algorithm**

I. Introduction

Recently, cyber-physical systems (CPSs) have increasingly been applied across various fields of modern society, such as smart grids [1], robotic systems [2], intelligent transportation systems [3], and electro-mechanical systems [4]. By deeply integrating physical devices with computing systems, these systems enable efficient sensing, communication, and control functions, significantly advancing productivity and societal progress. However, the complexity and openness of CPS make it vulnerable to faults [5], [6] and external attacks [7]. Ensuring its security and robustness has become a critical issue that must be urgently addressed.

Generally, network attacks can be broadly categorized into two types: deception attacks [8], [9] and denial-of-service (DoS) attacks [10]. The former achieves its objective by modifying specific data values, while the latter seeks to prevent information transmission. In recent years, many researchers have explored network security issues under deception and DoS attacks. For instance, [11] investigated recursive filters for random coupled complex networks affected by deception attacks and event-triggered random access schemes. In addition, adversarial agents pose a significant threat to CPS security and can be categorized into three types based on their behavioral characteristics: crash model, malicious model, and Byzantine model. Crashed agents exhibit functional failure, while malicious and Byzantine agents actively disrupt system operations. Specifically, malicious agents send consistent but incorrect values to all neighbors, while Byzantine agents are more adversarial, sending inconsistent values to different neighbors [12], [13]. Due to the complexity of Byzantine agents and their severe threat to the system, many papers focus on consensus under the Byzantine model. For example, in smart grids and vehicular networks, Byzantine attacks can disrupt system consensus, resulting in severe consequences. In recent years, researchers have extensively studied the Byzantine model, aiming to design optimization algorithms that mitigate the impact of Byzantine attacks.

Byzantine-resilient optimization has been extensively studied in centralized networks, where a central node aggregates global information and applies various robust aggregation rules. For example, the coordinate median rule [14] and geometric median rule [15] robustly estimate the average gradient by taking the global median of local gradients. [16] proposed a probability-based optimal estimation method for multisensor systems, where measurements from some sensor channels can be arbitrarily manipulated by attackers. The attack scenario considered in [16] belongs to a type of Byzantine agent attacks, which is less destructive. Due to the widespread use of distributed systems [17], [18], distributed optimization has become a prominent research topic. [19] proposed a reputation-based consensus algorithm for multi-agent networks. In this method, each agent assigns a reputation value to its neighbors, enabling it to discard messages from those exhibiting anomalous behavior. Another distributed optimization approach enables agents to reach consensus by minimizing the finite-

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sum cost function via local communication and computation [20]–[22]. However, achieving robust performance becomes more challenging when global information is unavailable and Byzantine attacks are present.

To mitigate the influence of Byzantine agents, distributed optimization commonly employs the sorting and filtering method [23], [24]. This method effectively reduces the impact of adversarial agents by discarding extreme values (both maximum and minimum) from neighbors. For instance, [20] focused on scalar local cost functions to achieve Byzantine-resilient multiagent optimization. This approach offered valuable insights for applications involving general local cost functions. [23] proposed a consensus-based distributed optimization algorithm to mitigate adversaries in the network topology by removing extreme values under certain conditions. However, the existing one-hop communication model has strict requirements on network structure and cannot accommodate more complex topologies. Therefore, the multi-hop relay-based approach is proposed in [25], [26] to enhance the network's fault tolerance and robustness by increasing the number of communication paths between agents. For instance, [27] tackled the Byzantine consensus problem under asynchronous updates and communication delays using the multi-hop weighted mean subsequence reduced algorithm. This algorithm not only reduced the network's connectivity requirements but also exhibited strong resilience to attacks in Byzantine consensus scenarios. Furthermore, multi-hop communication significantly improves the effectiveness of sorting and filtering methods, especially in scenarios with unreliable inter-node communication links. The multi-hop relay-based approach enables agents to achieve greater information redundancy, ensuring system consensus and stability even under adverse conditions.

Specifically, due to the widespread use of distributed systems, researchers focus on solving the distributed optimization-based secure state estimation problem for CPSs affected by adversarial agents [28]. For instance, [29] studied the distributed secure state estimation (DSSE) problem in CPSs monitored by a multi-agent network where some agents are adversarial. The paper transformed the distributed state estimation problem into a distributed optimization problem and proposed a distributed gradient descent algorithm. The core idea is to reduce the impact of adversarial agents by discarding extreme elements using sorting and filtering methods. However, under the Byzantine model, existing DSSE methods based on sorting-filtering technique mostly use one-hop communication, while the multi-hop relay-based method has rarely been explored for this purpose. Therefore, designing an efficient DSSE method that mitigates high computational complexity remains a significant challenge, which serves as the primary motivation for this study.

This paper focuses on the DSSE problem for CPSs, where Byzantine agents are present in a multi-agent network. The objective is to design a polynomial-time estimation strategy by integrating the multi-hop method into the filtering-sorting technique and derive a more general robustness condition. The key innovations of this paper are outlined as below:

1) This paper investigates polynomial-time secure state estimation through a novel multi-hop weighted distributed gradient descent (MW-DGD) algorithm. The key innovation in the proposed state estimation update strategy is the design of the mathematical equivalent representation under multi-hop communication. Different from existing methods, our approach integrates the sorting and filtering method with the minimum message coverage (MMC) to ensure that any adversarial value can be decomposed into a convex combination of two normal values.
2) A more general robustness condition to achieve consensus is established. Specifically, Theorem 3 provides the necessary and sufficient condition for achieving robust consensus under f-local Byzantine agents: the graph $\mathcal{G}$ is $(f+1)$-strictly robust with l hops. Compared to the $(2f+1)$-robustness condition for f-local Byzantine model in Theorem 2 and [23], Theorem 3 relaxes the graph robustness requirements. This is because Theorem 3 does not require each iteration to be rooted; instead, it provides the necessary and sufficient condition through interval contraction and extreme value boundedness.
3) A more universal distributed error detection (DED) algorithm is introduced to dynamically evaluate the estimation performance under Byzantine agents in multi-hop communication. Unlike existing methods, it incorporates the MMC concept, making it suitable for performance evaluation in broader multi-hop communication.

The rest of this paper is organized as follows. The system description and problem description are presented in Section II. Main results are shown in Sections III-VI, where Section III provides the MW-DGD algorithm, Section IV focuses on the consensus analysis, Section V discusses the convergence of the state estimation, and Section VI describes the DED algorithm. Furthermore, Section VII provides the simulation example. Finally, Section VIII concludes this paper.

Notation. For a matrix $G \in \mathbb{R}^{n_1 \times n_2}$, $[G]^i_j$ denotes the element in the ith row and jth column, such that $G = [g_{ij}]$ implies $[G]^i_j = g_{ij}$. G^T represents the transpose of G. $\lambda_M(G)$ and $\lambda_m(G)$ are the largest and smallest eigenvalues of G, separately. When $n_1 = n_2$, $He(G) = G + G^T$. The Kronecker product is denoted by $\otimes$. For two sets G' and G'', $G' \backslash G''$ is the complement of G'' in G', and $|G'|$ describes the cardinality of G'. $\mathbb{N}$ is a set of natural numbers. $1_M = [\begin{array}{ccc} 1 & \cdots & 1 \end{array}]^T \in \mathbb{R}^M$. Additionally, $I_{\{m\}}$ is a $1 \times n$ row vector with the m-th element equal to 1 and all other elements equal to 0.

II. Preliminaries

A. System Description

Consider a kind of linear CPSs in discrete form guided by a multi-agent network with N agents.

$$\begin{cases} x(t+1) = Ax(t) + Bu(t) + Dd(t), \\ y_i(t) = C_i x(t), i \in \mathcal{V} = \{1, \cdots, N\}, \end{cases} \tag{1}$$

in which $x(t) \in \mathbb{R}^n$ represents the system state with $\|x(t)\| < \infty$, $u(t) \in \mathbb{R}^{n_u}$ is the control input, $d(t) \in \mathbb{R}^{n_d}$ is the disturbance with $||d(t)|| \leq d_M$ (d_M is the constant characterizing the bound of noises), $y_i(t) \in \mathbb{R}^{n_i}$ is the system output.

A, B, D and C_i stand for the known parameters with proper dimensions.

Multi-agent network: It can be represented as the time-invariant directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = 1, \ldots, N$ is the set of agents and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of directed edges. Specially, the agents in $\mathcal{V}$ are partitioned into two subsets: normal agents $\mathcal{Q}$ and adversarial agents $\mathcal{A}$. Additionally, $\mathbb{A} = |\mathcal{A}|$ and $\mathbb{Q} = |\mathcal{Q}|$ stand for the number of adversarial and normal agents, respectively. The division between the adversarial and normal agents is unknown. An edge $(j, i) \in \mathcal{E}$ means that agent i can receive message from agent j. A l-hop path p from agent i_1 to agent i_l is a sequence of distinct agents $p = (i_1, i_2, \ldots, i_l)$ such that $(i_\hbar, i_{\hbar+1}) \in \mathcal{E}$ for all $0 \leq \hbar \leq l - 1$, characterizing multi-hop message propagation. For any agent $i \in \mathcal{V}$, its l-hop in-neighbor set $\mathcal{N}_i^{l-}$ is the set of agents that can reach agent i through no more than l hops. Likewise, the l-hop out-neighbor set $\mathcal{N}_i^{l+}$ represents the set of agents that originates from agent i to other agents after through at most l hops. For simplicity, when $l = 1$, we simplify $\mathcal{N}_i^{l-}$ ($\mathcal{N}_i^{l+}$) as $\mathcal{N}_i^{-}$ ($\mathcal{N}_i^{+}$).

Definition 1. (Byzantine agents [30]) An adversary agent $i \in \mathcal{A} \subseteq \mathcal{V}$ is called a Byzantine agent if the agent can change relay values and its own value, then transmit various relay values and state values to the neighbors.

Definition 2. (f-local/f-total set [31]) The set of adversary agents $\mathcal{A}$ is an f-local set when $|\mathcal{N}_i^{l-} \cap \mathcal{A}| \leq f$, for all normal agent $i \in \mathcal{Q}$. Similarly, $\mathcal{A}$ is an f-total set when $|\mathcal{A}| \leq f$.

Assumption 1. Globally, the set of adversarial agents is at most an s-total set. Locally, for all $i \in \mathcal{Q}$, the set of adversarial agents is an f-local set, where $f \in [0, s]$.

Definition 3. (s-sparse observable [32]) The CPS (1) is referred to be s-sparse observable if for any set $\mathcal{A} \subseteq \mathcal{V}$ with the condition $|\mathcal{A}| \leq s$, $(A, C_{\mathcal{Q}})$ can be observable. ($C_{\mathcal{Q}}$ consists of all rows from the matrices C_i where $i \in \mathcal{Q} = \mathcal{V} \backslash \mathcal{A}$.)

Assumption 2. The CPS (1) is $2s$-sparse observable.

Remark 1. Due to resource constraints, adversaries typically cannot compromise all agents. Consequently, Assumption 1 is introduced, similar to the sparsity constraint on attacks in [32]. Additionally, inspired by Theorem 3.2 in [32], CPS (1) is required to be $2s$-sparse observable for s-sparse sensor attacks, leading to Assumption 2.

B. Preliminaries

This subsection provides several definitions and lemmas that will be crucial to subsequent analysis.

Definition 4. ((r_1, r_2)-robust with l hops [27]) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is (r_1, r_2)-robust with l hops concerning any set $\mathcal{F} \subset \mathcal{V}$, if for each two disjoint nonempty subsets $\mathcal{V}_i \subset \mathcal{V}, i = 1, 2$, more than one of the following three requirements is satisfied:

(i) $\mathcal{Z}_{\mathcal{V}_1}^{r_1} = \mathcal{V}_1$, (ii) $\mathcal{Z}_{\mathcal{V}_2}^{r_1} = \mathcal{V}_2$, (iii) $|\mathcal{Z}_{\mathcal{V}_1}^{r_1}| + |\mathcal{Z}_{\mathcal{V}_2}^{r_1}| \geq r_2$, where $\mathcal{Z}_{\mathcal{V}_a}^{r_1}$ denotes the set of agents in $\mathcal{V}_a (a = 1, 2)$ that have no fewer than r_1 independent paths under no more than l hops from agents outside of $\mathcal{V}_a$, and none of these paths passing through agents in the set $\mathcal{F}$ as intermediate agents. Furthermore, if $\mathcal{G}$ meets this requirement w.r.t the subset $\mathcal{F}$ that satisfies the f-local model, then $\mathcal{G}$ is said to be (r_1, r_2)-robust with l hops under f-local model. When $r_2 = 1$, graph $\mathcal{G}$ becomes r_1-robust with l hops.

Definition 5. (r-strictly robust with l hops [27]) Assume that subset $\mathcal{F} \subset \mathcal{V}$ is a set of adversarial agents and let $\mathcal{G}_{\mathcal{H}}$ be the subgraph of $\mathcal{G}$ acquired by the agent set $\mathcal{H} = \mathcal{V} \backslash \mathcal{F}$. If $\mathcal{G}_{\mathcal{H}}$ is r-robust with l hops concerning $\mathcal{F}$ in $\mathcal{G}$, then $\mathcal{G}$ becomes r-strictly robust with l hops in the presence of f-local/total Byzantine model.

Definition 6. ((r_1, r_2)-resilient robust with l hops [29]) For $r_1, r_2 \in \mathbb{N}$, a graph $\mathcal{G}$ consisting of N agents is (r_1, r_2)-resilient robust with l hops if for any $\mathcal{G}_1$ acquired by deleting $r_1 - 1$ or fewer independent incoming edges[1] via at most l hops from any agent in $\mathcal{G}$, any agent in $\mathcal{G}_1$ still has no fewer than $N - r_2$ independent paths via no more than l hops from other agents to reach itself.

Definition 7. (minimum message coverage [26]) Consider the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, in which $\mathcal{S}$ denotes the set of transmissions sent within $\mathcal{G}$, and $\mathcal{P}(\mathcal{S})$ represents all paths corresponding to the transmissions in $\mathcal{S}$, i.e. $\mathcal{P}(\mathcal{S}) = \{path(s) : s \in \mathcal{S}\}$. The message coverage of $\mathcal{S}$ refers to the set of agents $\mathcal{T}(\mathcal{S}) \subset \mathcal{V}$ whose removal disrupts all paths, namely, for any path $P \in \mathcal{P}(\mathcal{S})$, one obtains $\mathcal{V}(P) \cap \mathcal{T}(\mathcal{S}) \neq \emptyset$. Specifically, the minimum message coverage of $\mathcal{S}$ is

$$\mathcal{T}^*(\mathcal{S}) \in \underset{\mathcal{T}(\mathcal{S}):\ Cover\ of\ \mathcal{S}}{\arg\min} |\mathcal{T}(\mathcal{S})| .$$

Remark 2. Definitions 4, 5, and 6 describe different robustness concepts. Definition 4 introduces the concept of r_1-robust graphs with l hops, which is applied in the proof of Theorem 2. Definition 5, describing r-strictly robust graphs with l hops, is used in the proof of Theorem 3 and imposes stricter conditions than Definition 4. Definition 6 defines (r_1, r_2)-resilient robust graphs with l hops, utilized in the proof of Theorem 4. Finally, Definition 7 introduces the concept of minimum message coverage, applied in the design of Algorithm 1.

Lemma 1. [23] Suppose the graph $\mathcal{G}$ is r-robust with l hops. The graph $\mathcal{G}'$ is constructed by removing up to $r - 1$ independent paths, each consisting of no more than l hops originating from agents in $\mathcal{G}$. Then, $\mathcal{G}'$ is rooted.

Lemma 2. For the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, let $L_k = [a_{ij,k}]$ be the weight matrix with $a_{ij,k} = \sum_{p \in \mathcal{P}} a_{ij,k}^p$, where the coefficient $a_{ij,k}^p > 0$ if and only if $p \in \mathcal{P}_{ij,k}$; otherwise, $a_{ij,k}^p = 0$. Here, the set $\mathcal{P}_{ij,k}$ denotes the retained, distinct paths from agent j to agent i at iteration k (each of length at most l).

When every agent in $\mathcal{G}$ contains no fewer than $N - s$ independent paths through at most l hops from other agents to itself, for $N' \geq \max\{0, N - s - 2\}$, every row of $\Phi_{k,k+N'} = L_{k+N'} \cdots L_k$ has no fewer than $N - s$ positive elements.

Proof: Referring to Lemma 2.7 in [29], suppose every agent in the graph $\mathcal{G}$ has at least $N - s$ independent paths from other agents to itself through l hops, and consider a matrix L_k' consisting of any $w < N - s$ rows of L_k:

If L_k' includes only w non-zero columns, then each agent $i \in \{i_1, \cdots, i_l\}$ lacks the path from agents outside $\{i_1, \cdots, i_l\}$ to agent i, contradicting the claim that every agent in $\mathcal{G}$ possesses at least $N - s$ independent paths from other agents to itself

[1]Independent incoming edges are the edges from other agents in the graph $\mathcal{G}$ pointing to the target agent i, where only agent i is common.

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through l hops. Consequently, L_k' must contain at least $w+1$ non-zero columns.

Let $\nu_{i,k,k+N'}$ represent the i-th row of $\Phi_{k,k+N'}$, and if the elements in L_k are non-negative, we have:

(i) $\nu_{i,k+N',k+N'}$ includes no fewer than two positive elements.

(ii) $\nu_{i,k+N'-1,k+N'} = \nu_{i,k+N',k+N'} L_{k+N'-1}$ includes no fewer than three positive elements.

(iii) $\nu_{i,k',k+N'} = \nu_{i,k'+1,k+N'} L_{k'}$ includes no fewer than $k+N'-k'+2$ positive elements.

By induction, $\nu_{i,k,k+N'}$ includes no fewer than $N-s$ positive elements ($\max\{0, N-s-2\} \leq N'$). ■

Lemma 1 discusses rootedness and will be applied in the proof of Theorem 2. Lemma 2 examines the number of positive elements in a matrix row and will be utilized in the proof of Theorem 4.

C. Problem Description

Distributed secure state estimation: For CPS (1), the multi-agent network $\mathcal{G}$ and the measurement output $y_i(t)$ corresponding to each agent, the goal is to develop DSSE methods to ensure that all normal agents in $\mathcal{Q}$ can obtain reliable state estimates, even when Byzantine agents in $\mathcal{A}$ satisfy Assumption 1:

$$\|x(t) - \hat{x}_i(t)\| \leq \theta\left(d_M, \bar{e}_0\right), i \in \mathcal{Q}, \tag{2}$$

where $\hat{x}_i(t)$ denotes the estimate value of $x(t)$ acquired by the agent i. Furthermore, $\theta\left(d_M, \bar{e}_0\right)$ is a bounded function regarding the upper bound of the disturbance d_M and the error $\bar{e}_0 = \max\limits_{i\in\mathcal{Q}} \|x(0) - \hat{x}_i(0)\|$.

As shown in Fig. 1, a framework of the attacked CPS and the multi-hop communication approach are given. In the distributed system considered in this paper, Byzantine agents directly interfere with the operation of the normal agents. We assume that each Byzantine agent can modify its own state and the values of the messages they transmit or forward.

Remark 3. Recently, the integration of multi-hop relay communication with traditional cellular networks and cognitive radio technology has emerged as a key research focus [33], [34]. In addition, the multi-hop relay-based method exhibits distinct advantages in smart grids with hierarchical substations (it allows data to be aggregated and forwarded through intermediate substations, significantly expanding the coverage of the monitoring system and creating redundant paths) and vehicular networks with V2V relays (it enables warning messages to propagate in a "hopping" fashion via intermediate vehicles, far exceeding the range of one-hop communication).

III. Distributed Secure State Estimation via a Multi-hop Weighted Gradient Descent Algorithm

For agent i, a series of n measurements, taken from t to $t+n-1$, are collected, yielding the following expression

$$\begin{aligned}\mathcal{Y}_i(t) &= \begin{bmatrix} y_i^T(t) & \cdots & y_i^T(t+n-1) \end{bmatrix}^T - F_i \mathcal{U}(t) \\ &= O_i x(t) + F_i \mathcal{D}(t), i \in \mathcal{Q},\end{aligned} \tag{3}$$

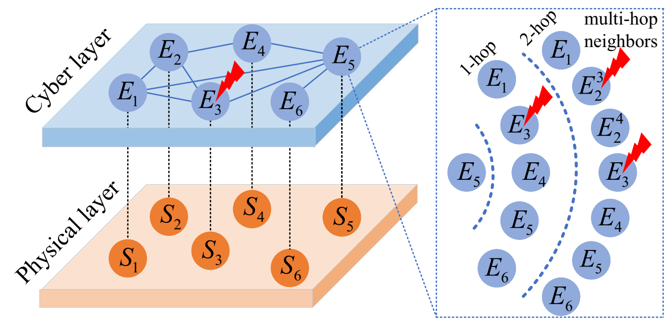

Fig. 1. The framework of the considered CPS. The physical layer has 6 local sensors (expressed as $S_i, i = 1, \cdots, 6$), the cyber layer is distributed with 6 estimated agents (expressed as $E_i, i = 1, \cdots, 6$). Each agent individually processes the measurements of its sensor and communicates with each other to reconstruct the system states. In a multi-hop relay network, take agent 5 as an example, its one-hop neighbors are $\mathcal{N}_5^{1-} = \{E_1, E_3, E_4, E_5, E_6\}$. Additionally, it receives relayed values from agent 2 via agents 3 and 4, represented by E_2^3 and E_2^4. Therefore, its 2-hop neighbors are $\mathcal{N}_5^{2-} = \{E_1, E_2^3, E_2^4, E_3, E_4, E_5, E_6\}$.

in which $\mathcal{U}(t) = \left[(Bu(t))^T, \cdots, (Bu(t+n-1))^T\right]^T$, $\mathcal{D}(t) = \left[(Dd(t))^T, \cdots, (Dd(t+n-1))^T\right]^T$, $||\mathcal{D}(t)|| \leq n^{\frac{1}{2}} d_M$,

$$F_i = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ C_i & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_i A^{n-2} & C_i A^{n-3} & \cdots & C_i \end{bmatrix},$$

$$O_i = \begin{bmatrix} C_i \\ C_i A \\ \vdots \\ C_i A^{n-1} \end{bmatrix}.$$

Referring to the DSSE problem discussed in Section II-C, as presented in [25], this can be transformed into a distributed optimization problem:

$$\begin{aligned} &\min_{\hat{x}_i \in R^n} \sum_{i\in\mathcal{Q}} ||\mathcal{Y}_i(t) - O_i \hat{x}_i||^2, \\ &s.t.\ \hat{x}_i = \hat{x}_j, i \neq j \in \mathcal{Q}. \end{aligned} \tag{4}$$

For an agent i, since it only collects $\mathcal{Y}_i(t)$, the global objective function $\sum_{i\in\mathcal{Q}} ||\mathcal{Y}_i(t) - O_i \hat{x}_i||^2$ is not available. Under Assumptions 1 and 2, $(A, C_\mathcal{Q})$ is observable, and taking $||\mathcal{D}(t)|| \leq n^{\frac{1}{2}} d_M$ into account, the solution $\hat{x} = \hat{x}_i = \hat{x}_j$ to the optimization problem (4) can be obtained satisfying

$$\|x(t) - \hat{x}\|^2 \leq \frac{1}{\lambda_{\min}\left(\sum_{i\in\mathcal{Q}} O_i^T O_i\right)} \sum_{i\in\mathcal{Q}} \|F_i(t)\|^2 n d_M^2, \tag{5}$$

which means that $\hat{x}$ is a reliable estimate of state $x(t)$.

If Byzantine agents are not present, optimization problems can be effectively solved using existing techniques [35]. However, when Byzantine agents are present, the set $\mathcal{Q}$ is typically unknown, preventing the direct solution of (4). In addition, the malicious model discussed in [36] represents a less severe threat than the Byzantine model, since malicious agents are required to send identical messages to all their neighbors. In the context of multi-hop communication, [27] considers the Byzantine model, but only for the one-dimensional case. Hence, the goal of this paper is to propose a secure strategy

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capable of effectively reconstructing the multi-dimensional states under Byzantine agents.

A. Update Rule

In this subsection, we define the update rule for the algorithm. At time t and iteration k, the normal agent i executes the following operations:

First, send the message $\hat{x}^{pm}_{ji,k} = [\hat{x}^{pm}_{i,k}, P^{m}_{ji,k}]$ consisting of the state $\hat{x}^{m}_{i,k}$ and corresponding path $P^{m}_{ji,k}$ to the agent $j \in \mathcal{N}^{l+}_{i}$, which is reached by at most l hops. To avoid any confusion, whenever variable $m \in \mathcal{M}_n = \{1, \cdots, n\}$ is introduced in optimization process or subsequent derivations, it refers to the specific index of the vector components, such as the m-th element of $\hat{x}_{i,k}$.

Then, agent i receives the message $\hat{x}^{pm}_{ij,k} = [\hat{x}^{pm}_{j,k}, P^{m}_{ij,k}]$ at most l hops from agent j. Specifically, $value[\hat{x}^{pm}_{ij,k}] = \hat{x}^{pm}_{j,k}$ is the path-dependent value originated at agent j and delivered to agent i and $path[\hat{x}^{pm}_{ij,k}] = P^{m}_{ij,k}$ is the route taken from j to i. Its primary role is to allow agent i to distinguish multiple values originating from the same source agent j but relayed through different paths.

Third, agent i updates itself as

$$\hat{x}^{m}_{i,k+1} = \sum_{j \in \hat{J}^{m}_{i,k}} \sum_{p \in \mathcal{P}^{m}_{ij,k}} a^{pm}_{ij,k} value[\hat{x}^{pm}_{ij,k}] + \eta_k I_{\{m\}} \Delta_{i,k}, \quad (6)$$

where $\mathcal{P}^{m}_{ij,k} = \left\{ path[\hat{x}^{pm}_{ij,k}] \middle| \hat{x}^{pm}_{ij,k} \in J^{m}_{i,k} \right\}$ is the set of paths from agent j to agent i, $a^{pm}_{ij,k} = 1/(|\hat{J}^{m}_{i,k}| \cdot |\mathcal{P}^{m}_{ij,k}|)$, $\Delta_{i,k} = O^{T}_{i} \left(\mathcal{Y}_i - O_i \sum_{m \in \mathcal{M}_n} I^{T}_{\{m\}} \bar{X}^{m}_{i,k} \right)$, $\bar{X}^{m}_{i,k} = \sum_{j \in \hat{J}^{m}_{i,k}} \sum_{p \in \mathcal{P}^{m}_{ij,k}} a^{pm}_{ij,k} value[\hat{x}^{pm}_{ij,k}]$. The path message also directly determines the weight assignment $a^{pm}_{ij,k}$. The assignment follows a principle: all retained messages share a total weight of 1. Thus, the weight is calculated as $a^{pm}_{ij,k} = 1/(|\hat{J}^{m}_{i,k}| \cdot |\mathcal{P}^{m}_{ij,k}|)$, where $|\hat{J}^{m}_{i,k}|$ is the number of unique source agents, $|\mathcal{P}^{m}_{ij,k}|$ is the number of distinct paths from agent j to i. This means that if an agent j has more paths to i, the weight assigned to each individual message originated by agent j is reduced proportionally.

Remark 4. The proposed algorithm does not require or assume consistency among values originating from the same source but traversing different paths. Regardless of whether multiple values from the same source are consistent, they are treated uniformly together with values from other sources and processed by the sorting and filtering method. Specifically, this method uses the MMC to identify and remove extreme values, thereby guaranteeing the convergence of the normal agents' states.

Remark 5. Every normal agent is aware of the value of f and the topology of $\mathcal{G}$ up to l hops. Here, f represents the pre-set theoretical upper bound on the number of Byzantine agents that the system can tolerate, rather than the real-time count of Byzantine agents during operation. The consensus of all normal agents can only be guaranteed when the actual number of Byzantine agents in the network does not exceed f. When the actual number of Byzantine agents exceeds f and these attackers are carefully designed, our algorithm cannot theoretically guarantee the consensus. Note that we assume that each Byzantine agent can modify its own state value $\hat{x}^{m}_{i,k}$ and the values of the messages it transmits or forwards. However, it cannot alter the path messages contained within those messages. In addition, the algorithm iteration is assumed to be synchronous.

B. Multi-hop Weighted Distributed Gradient Descent Algorithm

This subsection presents a DSSE method, where every agent has access to both its own state and that of its neighbors. During an attack, the Byzantine agents in $\mathcal{A}$ are unknown, presenting the primary challenge. Motivated by [27], we introduce the following MW-DGD algorithm.

Suppose Byzantine agents are restricted to an f-local set. Normal agents are unaware of which of their neighbors are Byzantine agents. Assume that on each iteration $k \in \mathbb{N}$, normal agent $i \in \mathcal{Q}$ performs Algorithm 1, which is illustrated in Fig. 2 and uses the notations defined in Table 1.

Algorithm 1 MW-DGD algorithm

1: At iteration k, normal agent i transmits its message $\hat{x}^{pm}_{ji,k} = [\hat{x}^{m}_{i,k}, P^{m}_{ji,k}]$ to each agent j in $\mathcal{N}^{l+}_{i}$. Next, agent i obtains the messages from the agents in $\mathcal{N}^{l-}_{i} \cup \{i\}$, denoted by $S^{m}_{i,k}$. Then, it sorts the messages in $S^{m}_{i,k}$.

2: (i) For each $m \in \mathcal{M}_n = \{1, \cdots, n\}$, define subsets as follows:

$$\bar{S}^{m}_{i,k} = \left\{ \hat{x}^{pm}_{ij,k} \in S^{m}_{i,k} : value(\hat{x}^{pm}_{ij,k}) > \hat{x}^{m}_{i,k} \right\},$$
$$\underline{S}^{m}_{i,k} = \left\{ \hat{x}^{pm}_{ij,k} \in S^{m}_{i,k} : value(\hat{x}^{pm}_{ij,k}) < \hat{x}^{m}_{i,k} \right\}.$$

(ii) If $\left| \mathcal{T}^{*}(\bar{S}^{m}_{i,k}) \right| \leq f$, we set $\bar{R}^{m}_{i,k} = \bar{S}^{m}_{i,k}$. Otherwise, let $\bar{R}^{m}_{i,k}$ be the largest subset of $\bar{S}^{m}_{i,k}$, such that (a) for all $\hat{x}^{1pm}_{ij,k} \in \bar{S}^{m}_{i,k} \backslash \bar{R}^{m}_{i,k}$ and $\hat{x}^{2pm}_{ij,k} \in \bar{R}^{m}_{i,k}$, we have $value(\hat{x}^{1pm}_{ij,k}) \leq value(\hat{x}^{2pm}_{ij,k})$, and (b) $\left| \mathcal{T}^{*}(\bar{R}^{m}_{i,k}) \right| = f$.

(iii) Define $\underline{R}^{m}_{i,k}$ from $\underline{S}^{m}_{i,k}$, which contains the smallest message values compared to $\hat{x}^{m}_{i,k}$.

(iv) Set $R^{m}_{i,k} = \bar{R}^{m}_{i,k} \bigcup \underline{R}^{m}_{i,k}$, $J^{m}_{i,k} = S^{m}_{i,k} \backslash R^{m}_{i,k}$ and $\hat{J}^{m}_{i,k}$, where $\hat{J}^{m}_{i,k}$ is the set of agents that produce the value in $J^{m}_{i,k}$.

3: Agent i updates its state estimate using Eq. (6).

4: Agent i transmits $\hat{x}_{i,k+1} = \sum_{m \in \mathcal{M}_n} I^{T}_{\{m\}} \hat{x}^{m}_{i,k+1}$ to agent $j \in \mathcal{V} \cap \mathcal{N}^{l+}_{i}$.

To provide a more detailed explanation of Algorithm 1, we focus on:

(i) Algorithm 1 is a locally distributed algorithm, such as agents only require local l-hops topology information rather than global information. Furthermore, because Algorithm 1 is an optimization method and does not rely on brute-force search, the proposed DSSE method achieves higher computational efficiency than existing methods. Specifically, for an NP-hard problem [16], [37], the brute-force search has a computational complexity of $O(N^s)$ (since C^{s}_{N} candidates are required). As N and s increase, the computational complexity grows exponentially. Unlike the brute-force search, the computational complexity in our method is derived as follows. Let

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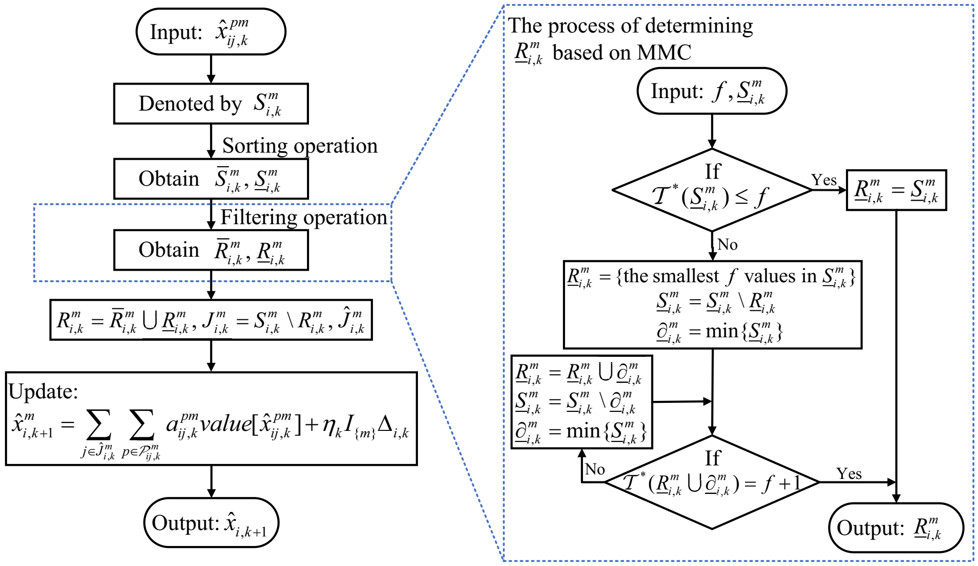

Fig. 2. The framework of the MW-DGD algorithm.

Table 1. Notations in MW-DGD algorithm.

Notation	Meaning
$S^m_{i,k}$	The set of all m-th component values collected by agent i from its l-hop in-neighborhood $\mathcal{N}^{l-}_i$. Here, values larger than $\hat{x}^m_{i,k}$ form $\bar{S}^m_{i,k}$ and values smaller than $\hat{x}^m_{i,k}$ form $\underline{S}^m_{i,k}$.
$R^m_{i,k}$	The set of values removed by the sorting and filtering method, with $R^m_{i,k} = \bar{R}^m_{i,k} \cup \underline{R}^m_{i,k}$. Here, $\bar{R}^m_{i,k}$ collects the trimmed largest values, and $\underline{R}^m_{i,k}$ collects the trimmed smallest values.
$J^m_{i,k}$	The set of retained values after filtering, i.e., $J^m_{i,k} = S^m_{i,k} \setminus R^m_{i,k}$.
$\hat{J}^m_{i,k}$	The set of source agents that generated the values contained in $J^m_{i,k}$.

$\mathbb{S}$ denote the number of candidate set $S^m_{i,k}$. The complexity of sorting operation is $O(\mathbb{S}\log\mathbb{S})$, and the complexity of filtering operation is $O(\mathbb{S})$. Therefore, the overall complexity of the sorting and filtering operation is $O(\mathbb{S}\log\mathbb{S})$. The analysis considers N sensors, iterations $k \in [1, K]$, and components $m \in \mathcal{M}_n = 1, \cdots, n$. Hence, Algorithm 1 is a polynomial-time algorithm with complexity $O(NKn\mathbb{S}\log\mathbb{S})$, i.e. $O(N)$ (since other parameters are fixed).

(ii) Algorithm 1 describes how normal agents in $\mathcal{Q}$ update their states, with each normal agent $i \in \mathcal{Q}$ executing Algorithm 1 collaboratively with other normal agents. Byzantine agents in $\mathcal{A}$, however, may never achieve reliable state estimates.

(iii) Because each agent forwards values from various neighbors, normal agent i can obtain multiple values from a single direct neighbor at each step. Obviously, as the number of hops l increases, the number of candidate agents grows, making the situation more complex. Therefore, the filtering operation acts on messages from the source agent j for all paths. To quantify the number of extreme values obtained by agent i from exactly f agents, the concept of MMC is introduced. This ensures the filtering mechanism remains effective even if Byzantine agents attempt to inject multiple adversarial values through multiple paths. Specifically, for agent i, $\bar{R}^m_{i,k}$ and $\underline{R}^m_{i,k}$ represent the largest message sets containing the maximum and minimum values, respectively, that could be generated or tampered with by f Byzantine agents. If the cardinality of the MMC of set $\underline{S}^m_{i,k}$ (in Step 2(i) of Algorithm 1) does not exceed f, agent i assigns $\underline{R}^m_{i,k} = \underline{S}^m_{i,k}$. Otherwise, agent i first assigns the largest f values from $\underline{S}^m_{i,k}$ to $\underline{R}^m_{i,k}$, then updates $\underline{S}^m_{i,k} = \underline{S}^m_{i,k} \backslash \underline{R}^m_{i,k}$, and defines $\underline{Q}^m_{i,k}$ as the minimum value in the updated set $\underline{S}^m_{i,k}$. If $|\mathcal{T}^*(\underline{R}^m_{i,k})| = f$, the sets $\underline{R}^m_{i,k} = \underline{R}^m_{i,k} \cup \underline{\partial}^m_{i,k}$, $\underline{S}^m_{i,k} = \underline{S}^m_{i,k} \backslash \underline{\partial}^m_{i,k}$, and the value $\underline{\partial}^m_{i,k} = \max\{\underline{S}^m_{i,k}\}$ are updated sequentially. This process continues until the MMC of $\underline{S}^m_{i,k}$ reaches $f+1$, yielding the final $\underline{R}^m_{i,k}$. After identifying the sets $\bar{R}^m_{i,k}$ and $\underline{R}^m_{i,k}$ in Step 2(iii) of Algorithm 1, agent i removes the values in set $R^m_{i,k} = \bar{R}^m_{i,k} \bigcup \underline{R}^m_{i,k}$ and updates the values using those in set $J^m_{i,k} = S^m_{i,k} \backslash R^m_{i,k}$.

Analyzing the update rule under (6) is challenging because $J^m_{i,k} \cap \mathcal{A}$ may contain elements, implying that $\hat{x}^m_{i,k}$ in (6) could originate from Byzantine agents. However, as extreme values are discarded in Step 2 of Algorithm 1 using the sorting and filtering method, the following theorem will be established. Theorem 1 demonstrates that agent state updates rely solely on the state estimates of normal agents.

Theorem 1. Given that Assumption 1 holds, each normal agent has at least $2f+1$ independent paths of at most l hops originating from agents in the set $\mathcal{N}^{l-}_i$. When $i \in \mathcal{Q}$, the update rule (6) is identical to

$$\hat{x}^m_{i,k+1} = \sum_{j\in\mathcal{N}^{l-}_i \cap \mathcal{Q}} \bar{a}^m_{ij,k} \hat{x}^m_{j,k} + \eta_k I_{\{m\}} \bar{\Delta}_{i,k}, \tag{7}$$

where $\bar{\Delta}_{i,k} = O^T_i \left(\mathcal{Y}_i - O_i \sum_{m\in\mathcal{M}_n} I^T_{\{m\}} \sum_{j\in N^{l-}_i \cap \mathcal{Q}} \bar{a}^m_{ij,k} \hat{x}^m_{j,k} \right)$, and the weights $\bar{a}^m_{ij,k}$ satisfy:

(i) $\sum_{j\in\mathcal{N}^{l-}_i \cap \mathcal{Q}} \bar{a}^m_{ij,k} = 1$,

(ii) $\bar{a}_{ii,k}^m \geq a$, $\bar{a}_{ij,k}^m = 0$ for $j \notin \mathcal{N}_i^- \cap \mathcal{Q}$, and at least $\mathcal{N}_i^- - 2f$ of the other weights $\bar{a}_{ij,k}^m \geq a/2$, where $a = 1 \Big/ \left[(|\mathcal{N}_i^{l-}|+1) \cdot \max\{|\mathcal{P}_{ij,k}^m|, 1\} \right] \in (0,1]$.

Proof: Consider a normal agent $i \in \mathcal{Q}$. We denote two distinct divisions for the in-neighbors of i. Regarding the first division, denote the sets $\bar{R}_{i,k}^m$, $\underline{R}_{i,k}^m$ and $J_{i,k}^m$, where $\bar{R}_{i,k}^m(\underline{R}_{i,k}^m)$ includes agents consisting of the maximum (minimum), which are deleted by agent i once the filtering strategy is applied. Additionally, we divide the set $J_{i,k}^m$ into two subsets: the neighbor agent set $J_{i,k}^{1m}$, which is directly connected to agent i, and non-neighbor agent set $J_{i,k}^{2m}$, which must be connected by $l-$ hop. For the second division, denote the sets $\bar{T}_{i,k}^m$, $\underline{T}_{i,k}^m$ and $\bar{J}_{i,k}^m$, where the minimum message cover of $\bar{T}_{i,k}^m(\underline{T}_{i,k}^m)$ containing the highest (lowest) values, equals f. The set $\bar{J}_{i,k}^m$ contains the remaining agents. Similarly, we can divide the set $\bar{J}_{i,k}^m$ into two subsets: the neighbor agent set $\bar{J}_{i,k}^{1m}$, which is directly connected to agent i, and non-neighbor agent set $\bar{J}_{i,k}^{2m}$, which must be connected by $l-$ hop. Therefore, we conclude that $\bar{R}_{i,k}^m \subseteq \bar{T}_{i,k}^m$, $\underline{R}_{i,k}^m \subseteq \underline{T}_{i,k}^m$, $\bar{J}_{i,k}^m \subseteq J_{i,k}^m$, $J_{i,k}^m = J_{i,k}^{1m} \cup J_{i,k}^{2m}$ and $\bar{J}_{i,k}^m = \bar{J}_{i,k}^{1m} \cup \bar{J}_{i,k}^{2m}$.

Case I: The agents in $\hat{J}_{i,k}^m$ have no adversarial values under different paths $\mathcal{P}_{ij,k}^m$

If there are no Byzantine agents and no relay values in $\hat{J}_{i,k}^m$ under different paths $\mathcal{P}_{ij,k}^m$, then $\hat{x}_{j,k}^{pm}$ is the same for state values from different paths for a fixed j. Therefore, we have $\bar{a}_{ii,k}^m = \sum_{p \in \mathcal{P}_{ii,k}^m} a_{ii,k}^{pm}$ and $\bar{a}_{ij,k}^m = \sum_{p \in \mathcal{P}_{ij,k}^m} a_{ij,k}^{pm}$ for $j \in \hat{J}_{i,k}^m$. Specifically,

$$\bar{a}_{ii,k}^m + \sum_{j \in \mathcal{N}_i^{l-} \cap \mathcal{Q}} \bar{a}_{ij,k}^m = \sum_{j \in \hat{J}_{i,k}^m} \sum_{p \in \mathcal{P}_{i,k}^m} a_{ij,k}^{pm} = 1,$$

where satisfies the first condition in the proposition. In addition, since $\left| J_{i,k}^{1m} \right| \geq \left| \mathcal{N}_i^- \right| - 2f$ and $0 < \left| \hat{J}_{i,k}^m \right| \leq \left| \mathcal{N}_i^{l-} \right| + 1$, at least $\left| \mathcal{N}_i^- \right| - 2f$ the weights are bounded below by $a = 1 \Big/ \left[(|N_i^{l-}|+1) \cdot \max\{|P_{ij,k}^m|, 1\} \right] \in (0,1]$, this meets condition (ii) in the proposition.

Case II: The agents in $\hat{J}_{i,k}^m$ have one or more adversarial values under different paths $\mathcal{P}_{ij,k}^m$

Now, consider the case where the agent in $\hat{J}_{i,k}^m$ has one or more adversarial values under different paths $\mathcal{P}_{ij,k}^m$. We consider the adversarial values caused by agents in $\hat{\bar{J}}_{i,k}^m$ and $\hat{J}_{i,k}^m \backslash \hat{\bar{J}}_{i,k}^m$, respectively. Sets $\hat{J}_{i,k}^m$ and $\hat{\bar{J}}_{i,k}^m$ contain values averaged from those obtained by the same agent through different paths in sets $J_{i,k}^m$ and $\bar{J}_{i,k}^m$, respectively. Consequently, all agent values in sets $\hat{J}_{i,k}^m$ and $\hat{\bar{J}}_{i,k}^m$, derived from different paths $\mathcal{P}_{ij,k}^m$, are included in sets $J_{i,k}^m$ and $\bar{J}_{i,k}^m$. Furthermore, two categories of adversarial values in sets $\bar{J}_{i,k}^m$ and $J_{i,k}^m \backslash \bar{J}_{i,k}^m$ can be considered as follows.

1) Consider the adversarial value $\hat{x}_{iz,k}^{pm} \in J_{i,k}^m \backslash \bar{J}_{i,k}^m$ that agent i received from agent z. Since agent i retained the value from agent z, either $\left| \mathcal{T}^* \left(\bar{R}_{i,k}^m \right) \right| = f$ or $value[\hat{x}_{iz,k}^{pm}] < \hat{x}_{i,k}^m$ must hold. Similarly, either $\left| \mathcal{T}^* \left(\underline{R}_{i,k}^m \right) \right| = f$ or $value[\hat{x}_{iz,k}^{pm}] > \hat{x}_{i,k}^m$ must hold. Hence, two cases arise: (i) $\left| \mathcal{T}^* \left(\bar{R}_{i,k}^m \right) \right| = f$ and $value[\hat{x}_{iz,k}^{pm}] > \hat{x}_{i,k}^m$; or (ii) $value[\hat{x}_{iz,k}^{pm}] < \hat{x}_{i,k}^m$ and $\left| \mathcal{T}^* \left(\underline{R}_{i,k}^m \right) \right| = f$.

We analyze Case (i) in two steps to demonstrate that the adversarial value $\hat{x}_{iz,k}^{pm}$ is bounded by normal values. **Step 1: Existence of a lower normal bound.** Since $value[\hat{x}_{iz,k}^{pm}] > \hat{x}_{i,k}^m$ and agent i is normal, $\hat{x}_{i,k}^m$ is a normal value that provides a lower bound. **Step 2: Existence of an upper normal bound.** Let $\bar{W}_{i,k}^m$ be the set of retained values no less than $\hat{x}_{iz,k}^{pm}$ (note that $\bar{W}_{i,k}^m \supset \bar{R}_{i,k}^m$). If all values in $\bar{W}_{i,k}^m$ are adversarial, then by $|\mathcal{T}^*(\bar{R}_{i,k}^m)| = f$ and the definition of MMC, we have $|\mathcal{T}^*(\bar{W}_{i,k}^m)| > f$. This means the values in $\bar{W}_{i,k}^m$ must be sent or relayed by more than f distinct Byzantine agents, which contradicts the f-local model. Therefore, $\bar{W}_{i,k}^m$ must contain at least one normal value. Furthermore, since $\hat{x}_{iz,k}^{pm}$ is adversarial, there must exist a normal value greater than $\hat{x}_{iz,k}^{pm}$. Combining Steps 1-2, $\hat{x}_{iz,k}^{pm}$ is guaranteed to lie between two normal values. Similarly, we obtain the same conclusion for Case (ii).

Given that there are at most f Byzantine agents in $\mathcal{N}_i^{l-}$, there exists a pair of normal agents $u, d \in \mathcal{N}_i^{l-} \cup \{i\}$ such that $value[\hat{x}_{id,k}^{p'm}] \leq value[\hat{x}_{iz,k}^{pm}] \leq value[\hat{x}_{iu,k}^{p''m}]$. Hence, $a_{iz,k}^{pm} value[\hat{x}_{iz,k}^{pm}]$ can be expressed as

$$\begin{aligned} a_{iz,k}^{pm} value[\hat{x}_{iz,k}^{pm}] =& a_{iz,k}^{pm}(1-\gamma_z^{pm}) value[\hat{x}_{iu,k}^{p'm}] \\ &+ a_{iz,k}^{pm}\gamma_z^{pm} value[\hat{x}_{id,k}^{p''m}], \gamma_z^{pm} \in [0,1]. \end{aligned}$$

Then, the weights $a_{id,k}^{p'm}$ and $a_{iu,k}^{p''m}$ are updated to $a_{id,k}^{p'm} \leftarrow a_{id,k}^{p'm} + a_{iz,k}^{pm}\gamma_z^{pm}$ and $a_{iu,k}^{p''m} \leftarrow a_{iu,k}^{p''m} + a_{iz,k}^{pm}(1-\gamma_z^{pm})$. Consequently, the adversarial value $\hat{x}_{iz,k}^{pm} \in J_{i,k}^m \backslash \bar{J}_{i,k}^m$ can be decomposed into contributions from the two normal values.

Next, considering the set $\bar{J}_{i,k}^{1m}$, we have $\left| \bar{J}_{i,k}^{1m} \right| = \left| \mathcal{N}_i^- \right| - 2f$. If $\bar{J}_{i,k}^m$ contains no adversarial values, the construction of $\bar{a}_{ij,k}^m = \sum_{p \in \mathcal{P}_{ij,k}^m} a_{ij,k}^{pm}$ is accomplished, and the two requirements stated in proposition are satisfied. This is because the weights allocated to the normal agents in $\bar{J}_{i,k}^m$ meet the requirement (ii), as $\bar{a}_{ij,k}^m \geq a$.

2) If the adversarial values lie in $\bar{J}_{i,k}^m$, then both $|\mathcal{T}^*(\bar{R}_{i,k}^m)| = f$ and $|\mathcal{T}^*(\underline{R}_{i,k}^m)| = f$ hold. Based on the analysis above, $\left| \mathcal{T}^* \left(\bar{R}_{i,k}^m \right) \right| = f$ implies the existence of a normal value greater than $\hat{x}_{iz,k}^{pm}$, and $\left| \mathcal{T}^* \left(\underline{R}_{i,k}^m \right) \right| = f$ implies the existence of a normal value less than it. Hence, $\hat{x}_{iz,k}^{pm}$ is again guaranteed to be bounded by normal values.

Assume that there are H adversarial values in $\bar{J}_{i,k}^m$, where $\bar{J}_{i,k}^{1m}$ contains $H_1 (0 \leq H_1 \leq f)$ adversarial values and $\bar{J}_{i,k}^{2m}$ contains $H - H_1$ adversarial values. Specially, all the adversarial values in $\bar{J}_{i,k}^{1m}$ come from the Byzantine agents themselves. According to the above analysis, there must be at least H_1 normal agents in both $\bar{T}_{i,k}^m$ and $\underline{T}_{i,k}^m$. Let the states of H_1 Byzantine agents in $\bar{J}_{i,k}^{1m}$ be denoted as

$$\left\{ value[\hat{x}_{iz_1,k}^{p_1 m}], value[\hat{x}_{iz_2,k}^{p_2 m}], \cdots, value[\hat{x}_{iz_{H_1},k}^{p_{H_1} m}] \right\}.$$

Select the states of H_1 normal agents in $\underline{T}_{i,k}^m$ as

$$\left\{ value[\hat{x}_{id_1,k}^{p'_1 m}], value[\hat{x}_{id_2,k}^{p'_2 m}], \cdots, value[\hat{x}_{id_{H_1},k}^{p'_{H_1} m}] \right\}$$

and the states of any H_1 normal agents in $\bar{T}_{i,k}^m$ as

$$\left\{value[\hat{x}_{iu_1,k}^{p_1''m}], value[\hat{x}_{iu_2,k}^{p_2''m}], \cdots, value[\hat{x}_{iu_{H_1},k}^{p_{H_1}''m}]\right\}.$$

First, consider $H_1 (0 \leq H_1 \leq f)$ adversarial values in $\bar{J}_{i,k}^{1m}$. For $1 \leq j \leq H_1$, we have $value[\hat{x}_{id_j,k}^{p_j'm}] \leq value[\hat{x}_{iz_j,k}^{p_jm}] \leq value[\hat{x}_{iu_j,k}^{p_j''m}]$. Thus, $value[\hat{x}_{iz_j,k}^{p_jm}]$ can be described as

$$\begin{aligned} value[\hat{x}_{iz_j,k}^{p_jm}] =& \gamma_j^{p_jm} value[\hat{x}_{id_j,k}^{p_j'm}] \\ &+ (1-\gamma_j^{p_jm}) value[\hat{x}_{iu_j,k}^{p_j''m}], \end{aligned}$$

where $0 \leq \gamma_j^{pm} \leq 1$. Note that the states of the Byzantine agents are a convex combination of the states of normal agents $value[\hat{x}_{id_j,k}^{p_j'm}]$ and $value[\hat{x}_{iu_j,k}^{p_j''m}]$. This implies either γ_j^{pm} or $1-\gamma_j^{pm}$ must be no less than 0.5. Update their weights $a_{id_j,k}^{p_j'm} \leftarrow a_{id_j,k}^{p_j'm} + a_{iz_j,k}^{p_jm}\gamma_j^{p_jm}$ and $a_{iu_j,k}^{p_j''m} \leftarrow a_{iu_j,k}^{p_j''m} + a_{iz_j,k}^{p_jm}(1-\gamma_j^{p_jm})$ for $1 \leq j \leq H_1$. Then, we consider $H - H_1$ adversarial values in $\bar{J}_{i,k}^{2m}$. Let the states of $H - H_1$ Byzantine agents in $\bar{J}_{i,k}^{2m}$ be denoted as

$$\left\{value[\hat{x}_{iz_1,k}^{p_1'''m}], value[\hat{x}_{iz_2,k}^{p_2'''m}], \cdots, value[\hat{x}_{iz_{H-H_1},k}^{p_{H-H_1}'''m}]\right\}.$$

Since the sets $\bar{T}_{i,k}^m$ and $\underline{T}_{i,k}^m$ are non-empty, the values in $\bar{J}_{i,k}^{2m}$ can be represented as a convex combination of the two normal agents in $\bar{T}_{i,k}^m$ and $\underline{T}_{i,k}^m$. Therefore, for Case I and Case II, condition (i) is satisfied.

Based on above discussion, $\bar{a}_{ij,k}^m \geq a_{ij,k}^{pm} \geq a$ for $\bar{J}_{i,k}^{1m} \cap \mathcal{U}$ guarantees that $\left|\bar{J}_{i,k}^{1m}\right| - H_1 = \left|\mathcal{N}_i^-\right| - 2f - H_1$ weights are lower bounded by a. Note that in case I, at least H_1 agents in $\{u_1, u_2, \cdots, u_{H_1}\} \cup \{d_1, d_2, \cdots, d_{H_1}\}$ are assigned weights that are lower bounded by $a/2$. Therefore, the distribution of the Byzantine agents weights ensures that the additional H_1 normal agents are assigned weights that are lower bounded by $a/2$. In conclusion, at least $\left|\mathcal{N}_i^-\right| - 2f$ weights (except for $\bar{a}_{ii,k}^m$) are lowed bounded by $a/2$. ■

Remark 6. The key idea of the proof for Theorem 1 is mainly divided into two cases: Case I: the agents in $\hat{J}_{i,k}^m$ have no adversarial values under different paths $\mathcal{P}_{ij,k}^m$, and Case II: the agents in $\hat{J}_{i,k}^m$ have one or more adversarial values under different paths $\mathcal{P}_{ij,k}^m$. In Case I, the analysis is straightforward. In Case II, our approach integrates the sorting and filtering method with MMC to ensure that any adversarial value during iteration can be decomposed into a convex combination of two normal values. Specifically, condition (i) is analyzed with l-hops neighbor agents, and condition (ii) is analyzed with one-hop neighbor agents. Due to the local filtering operation of each normal agent, dynamic (7) is mathematically equivalent to dynamic (6), which will play a key role in the proof in the next section.

IV. Consensus Analysis

Section IV discusses the consensus analysis of normal agents through the MW-DGD algorithm. The section is divided into two subsections. The first subsection provides the sufficient conditions to reach consensus under f-local Byzantine agents. The second subsection presents the necessary and sufficient conditions for consensus under f-local Byzantine agents. Compared to Theorem 2, Theorem 3 gives a less restrictive graph connectivity condition.

A. Sufficient Condition for Consensus under f-local Byzantine Agents

This subsection proves the sufficient condition for the consensus of algorithm under f-local Byzantine agents through being rooted at each iteration. Theorem 2 below generalizes the one-hop approach in Theorem 6.1 of [23].

Theorem 2. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent a directed graph with l-hop communication. Each normal agent updates its state using the MW-DGD algorithm with parameter f. If $\mathcal{G}$ is $(2f+1)$-robust with l hops, the subgradients are bounded by $G = \sup_{k\geq 0}\left|I_{\{m\}}\Delta_{i,k}\right|$, and $\lim_{k\to\infty}\eta_k = 0$, then there exists a row vector q_k^m such that

$$\lim_{k\to\infty}\left\|\hat{x}_{i,k}^m - \tilde{x}_k^m\right\| = 0, i \in \mathcal{Q}, \tag{8}$$

where $\tilde{x}_k^m = q_k^m \hat{X}_k^m$, $\hat{X}_k^m$ is a vector composed of $\hat{x}_{i,k}^m$, and $q_k^m = q_{k+1}^m \bar{L}_k^m$.

Proof: According to Theorem 1, we know that a is the lower bound of the diagonal elements for the matrix $\bar{L}_k^m = [\bar{a}_{ij,k}^m]$ in (7), and each row has at least $\left|\mathcal{N}_i^-\right| - 2f$ one-hop neighbors are lower bounded by $a/2$. In graph $\mathcal{G}$, delete edges with weights lower than $a/2$. This operation removes all edges between the agents corresponding to the adversarial values and normal agents. For each normal agent, the deleted adversarial values are distributed across at most $2f$ independent paths from other agents, as specified in Algorithm 1. Furthermore, consider that there are at least $\left|\mathcal{N}_i^-\right| - 2f$ one-hop neighbors, each lower-bounded by $a/2$. This implies that there are at least $\left|\mathcal{N}_i^-\right| - 2f$ independent paths from other agents. Based on Lemma 1, if the graph $\mathcal{G}$ is $(2f+1)$-robust with l hops, the subgraph of normal agents is rooted after removing the edges where the minimum cover of the normal agents is smaller than $2f$. In summary, $\bar{L}_k^m$ is rooted at each iteration, and all of its edge weights are lower-bounded by $a/2$. Referring to the first part of Lemma 3.4 in [23], condition (8) is derived and the theorem is proved. ■

Remark 7. Theorem 2 generalizes the one-hop problem and reduces to Theorem 6.1 in [23] when $l = 1$. The preceding proof is based on the property that in a $(2f+1)$-robust with l-hop network, the matrix $\bar{L}_k^m$ for a normal agent is rooted within the f-local Byzantine model. However, this serves only as a sufficient condition and imposes strict requirements on graph connectivity. Therefore, the following subsection presents a necessary and sufficient condition for the agents consensus while relaxing the graph connectivity constraints.

Remark 8. For any $k \geq s \geq 0$, let $\Phi_{k,s}^m = L_k^m \cdots L_s^m$. Under the condition that the matrices have a rooted subgraph and uniformly bounded positive diagonal and edge weights, for each fixed s, there exists a row-stochastic vector q_s^m such that $\lim_{k\to\infty}\Phi_{k,s}^m = \mathbf{1}q_s^m$ and $q_s^m = q_{s+1}^m L_s^m$. Consequently, define a row-stochastic vector q_k^m satisfying $q_k^m = q_{k+1}^m L_k^m$. Let $\tilde{x}_k^m \triangleq q_k^m \hat{X}_k^m$ and $\hat{X}_k^m = [\hat{x}_{1,k}^m, \ldots, \hat{x}_{\mathbb{Q},k}^m]^T$. Since q_k^m

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is row-stochastic, $\tilde{x}_k^m$ always lies within the convex hull of the normal agents' states and can be regarded as the consensus agent induced by Eq. (8). Under the graph and filtering conditions, it follows that $\|\hat{x}_{i,k}^m - \tilde{x}_k^m\| \to 0$ for all normal agents i, and thus $\tilde{x}_k^m$ shares the same limit as the all consensus value. In short, although q_k^m evolves over time, it characterizes the effective influence weights, making $\tilde{x}_k^m$ a metric for measuring and tracking consensus.

B. Necessary and Sufficient Condition for Consensus under f-local Byzantine Agents

If the graph fails to be $(2f+1)$-robust with l hops, the network generated through the filtering operation might not remain rooted at each iteration. Inspired by [27], this section establishes the necessary and sufficient condition for achieving consensus under Byzantine agents. To prove this result, we define the following variables:

Let $\hat{\bar{x}}_k^m = \max\limits_{i\in\mathcal{Q}} \hat{x}_{i,k}^m$, $\hat{\underline{x}}_k^m = \min\limits_{i\in\mathcal{Q}} \hat{x}_{i,k}^m$ and $D_k^m = \hat{\bar{x}}_k^m - \hat{\underline{x}}_k^m$. For $\forall k \in \mathbb{N}$, we have $\delta_k = \sup_{\bar{k}\geq k} |\eta_{\bar{k}}| G$, where $G = \sup_{k\geq 0} \left| I_{\{m\}} \Delta_{i,k} \right|$ represents the upper bound of the subgradients. Obviously, $\left|\eta_{\bar{k}} I_{\{m\}} \Delta_{i,\bar{k}}\right| \leq \delta_k$ for $\bar{k} \geq k$. Denote the following sets

$$Z_1^m(\bar{k},\gamma) = \left\{ j \in \mathcal{Q} | \hat{x}_{j,\bar{k}}^m > \hat{\bar{x}}_k^m - \gamma \right\},$$

$$Z_2^m(\bar{k},\gamma) = \left\{ j \in \mathcal{Q} | \hat{x}_{j,\bar{k}}^m < \hat{\underline{x}}_k^m + \gamma \right\}.$$

Theorem 3. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a directed graph with l-hop communication. Each normal agent updates its state via a MW-DGD algorithm with parameter f. Resilient asymptotic consensus can be achieved in the presence of f-local Byzantine adversaries if and only if $\mathcal{G}$ is called to be $(f+1)$-strictly robust with l hops.

Proof: (Necessity) The necessity can be obtained from Proposition 10 in [27].

(Sufficiency) Define $\gamma_0 = D_k^m/2$. Note that the sets $Z_1^m(k,\gamma_0)$ and $Z_2^m(k,\gamma_0)$ are mutually disjoint. According to the definition of $\hat{\bar{x}}_k^m$, at least one normal agents is included in the set $Z_1^m(k+\tau,\gamma_\tau)$ for $\tau = 0, 1, \cdots, |\mathcal{Q}|$. We assume that the set $Z_1^m(k,\gamma_0)$ is nonempty. To prove this, we must show that if a normal agent j does not belong to the set $Z_1^m(k+\tau,\gamma_\tau)$, it also does not belong to the set $Z_1^m(k+\tau+1,\gamma_{\tau+1})$. Assuming agent j satisfies $\hat{x}_{j,k}^m \leq \hat{\bar{x}}_k^m - \gamma_0$, the value of agent j at iteration $k+1$ is bounded as

$$\begin{aligned} \hat{x}_{j,k+1}^m &\leq (1-a)\,\hat{\bar{x}}_k^m + a\left(\hat{\bar{x}}_k^m - \gamma_0\right) + \delta_k \\ &= \hat{\bar{x}}_k^m - a\gamma_0 + \delta_k. \end{aligned} \tag{9}$$

Let $\gamma_\tau = a^\tau \gamma_0 - \tau\delta_k$. It follows that $\hat{x}_{j,k+1}^m \leq \hat{\bar{x}}_k^m - \gamma_1$. Therefore, the cardinality of the set $Z_1^m(\bar{k},\gamma)$ is nonincreasing, which means that the set $Z_1^m(k,\gamma_0)$ is nonempty. Similarly, the set $Z_2^m(k,\gamma_0)$ is nonempty.

Next, we will demonstrate that one of these two sets becomes empty within finite time. Since the network $\mathcal{G}$ is $(f+1)$-strictly robust with l hops, $\mathcal{G}_\mathcal{Q} = (\mathcal{V}, \mathcal{E}_\mathcal{Q})$, where $\mathcal{E}_\mathcal{Q}$ denotes the set of directed edges among normal agents, is $(f+1)$-robust with l hops for the set $\mathcal{A}$. Hence, between the two nonempty, disjoint sets $Z_1^m(k,\gamma_0)$ and $Z_2^m(k,\gamma_0)$, at least one contains a normal agent with $f+1$ or more independent normal paths originating from normal agents outside.

Assume that the normal agent $i \in Z_1^m(k,\gamma_0)$ has no less than $f+1$ independent paths originating from normal agents outside. Based on the above analysis, these normal agents cannot belong to $Z_1^m(k+\tau,\gamma_\tau)$. According to the Algorithm 1, there must be agents involved in the update process, as agent i can only remove the smallest value whose the minimum message cover is f. Therefore, the value of agent i at the next iteration is bounded with

$$\begin{aligned} \hat{x}_{i,k+1}^m &\leq \hat{\bar{x}}_k^m - a\gamma_0 + \delta_k \\ &\leq \hat{\bar{x}}_k^m - \gamma_1. \end{aligned} \tag{10}$$

Likewise, after $|\mathcal{Q}|$ iterations, we have

$$\hat{x}_{i,k+|\mathcal{Q}|}^m \leq \hat{\bar{x}}_k^m - \gamma_{|\mathcal{Q}|}.$$

which implies that at least one of sets $Z_1^m(k+|\mathcal{Q}|,\gamma_{|\mathcal{Q}|})$ or $Z_2^m(k+|\mathcal{Q}|,\gamma_{|\mathcal{Q}|})$ becomes empty after $|\mathcal{Q}|$ iterations. Assume that the former set becomes empty, this implies that

$$\hat{\bar{x}}_{k+|\mathcal{Q}|}^m \leq \hat{\bar{x}}_k^m - \gamma_{|\mathcal{Q}|}.$$

Similarly, we obtain $\hat{\underline{x}}_{k+|\mathcal{Q}|}^m \geq \hat{\underline{x}}_k^m - |\mathcal{Q}|\delta_k$, thus

$$\begin{aligned} D_{k+|\mathcal{Q}|}^m &\leq D_k^m - \gamma_{|\mathcal{Q}|} + |\mathcal{Q}|\delta_k \\ &= \left(1 - \frac{a^{|\mathcal{Q}|}}{2}\right) D_k^m + 2|\mathcal{Q}|\delta_k. \end{aligned}$$

Fix $\forall k \in \mathbb{N}$, for $r \in \mathbb{N}$ and $|\mathcal{Q}| = \mathbb{Q}$, we acquire that

$$D_{k+r\mathbb{Q}}^m \leq \tilde{a}^r D_k^m + 2\mathbb{Q}\sum_{h=0}^{r-1} \tilde{a}^{r-1-h}\delta_{k+h\mathbb{Q}}, \tag{11}$$

where $\mathbb{Q}$ and $\delta_{k+h\mathbb{Q}}$ are an unknown but bounded constant (since $\left|I_{\{m\}}\Delta_{i,k}\right|$ is bounded), and $\tilde{a} = 1 - a^{\mathbb{Q}}/2$.

If $\lim_{k\to\infty}\eta_k = 0$, we obtain $\lim_{k\to\infty}\delta_k = 0$, which means that $\lim_{h\to\infty}\delta_{k+h\mathbb{Q}} = 0$. Hence, $\lim_{r\to\infty} D_{k+r\mathbb{Q}}^m = 0$. Since the above condition holds for $\forall k \in \mathbb{N}$, we have $\lim_{k\to\infty} D_k^m = 0$. Therefore, we acquire $\lim_{k\to\infty}\hat{\bar{x}}_k^m = \lim_{k\to\infty}\hat{\underline{x}}_k^m$. ■

Remark 9. Unlike Theorem 2, which requires guaranteed rootedness, the key to proving Theorem 3 is that, in any iteration, no normal agent uses a value outside the range of $[\hat{\underline{x}}_k^m, \hat{\bar{x}}_k^m]$ in its update formula (6). The specific analysis is as follows. Consider the value set $J_{i,k}^m$, where values are not filtered out based at the iteration k. If the set includes only normal agents, their values will clearly fall within the interval $[\hat{\underline{x}}_k^m, \hat{\bar{x}}_k^m]$. On the other hand, suppose $J_{i,k}^m$ contains H adversarial values. Since agent i removes the most outlier value in the neighborhood during iteration k, the value of these H adversarial values must be medium compared to the removed value. In the f-local Byzantine set, at least one normal agent within the neighborhood of agent i has a value exceeding those of H adversarial values. Similarly, at least one normal agent has a value less than those of H adversarial values (this normal agent can be agent i itself). Therefore, all values used by the agent i during iteration k fall within the interval $[\hat{\underline{x}}_k^m, \hat{\bar{x}}_k^m]$.

Remark 10. Proposition 11 in [27] implies that $\mathcal{G}$ is named to be $(f+1)$-strictly robust with l hops if the graph condition

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$\mathcal{G}$ is $(2f+1)$-robust with l hops. Compared to the $(2f+1)$-robustness condition under f-local adversaries in [23], [29], we derive a more relaxed graph condition. Hence, different from Theorem 2, Theorem 3 provides a less restrictive necessary and sufficient condition on graph connectivity requirements to ensure consensus.

V. Convergence Analysis

This section focuses on analyzing the convergence of the acquired state estimation. Lemma 3 examines the consensus for the state values of normal agents, while Lemma 4 evaluates the reliability for the acquired state estimation.

For the convenience of subsequent proof, the following notations are introduced.

$$\hat{X}_k^m = \begin{bmatrix} \hat{x}_{i_1,k}^m \\ \vdots \\ \hat{x}_{i_{\mathbb{Q}},k}^m \end{bmatrix}, \hat{X}_k = \begin{bmatrix} \hat{X}_k^1 \\ \vdots \\ \hat{X}_k^n \end{bmatrix} \in \mathbb{R}^{n\mathbb{Q}},$$

$$\tilde{x}_k = q_k \hat{X}_k = \begin{bmatrix} \tilde{x}_k^1 \\ \vdots \\ \tilde{x}_k^n \end{bmatrix} \in \mathbb{R}^n, F = \begin{bmatrix} F_{i_1} \\ \vdots \\ F_{i_{\mathbb{Q}}} \end{bmatrix},$$

$$\mathcal{Y} = \begin{bmatrix} \mathcal{Y}_{i_1}^T & \cdots & \mathcal{Y}_{i_{\mathbb{Q}}}^T \end{bmatrix}^T = O\left(1_{\mathbb{Q}} \otimes I_n\right) x + F\mathcal{D},$$
$$O = diag\left\{ \begin{matrix} O_{i_1} & \cdots & O_{i_{\mathbb{Q}}} \end{matrix} \right\},$$
$$q_k = diag\left\{ \begin{matrix} q_k^1 & \cdots & q_k^n \end{matrix} \right\} \in \mathbb{R}^{n\mathbb{Q}\times n\mathbb{Q}},$$
$$\bar{L}_k = diag\left\{ \begin{matrix} \bar{L}_k^1 & \cdots & \bar{L}_k^n \end{matrix} \right\} \in \mathbb{R}^{n\mathbb{Q}\times n\mathbb{Q}},$$

where $i_1, i_2, \cdots, i_{\mathbb{Q}}$ are the indices of all normal agents in the set $\mathcal{Q}$, and their state estimates collectively form the vector $\hat{X}_k^m$, and $q_k = q_{k+1}\bar{L}_k$. In addition, we define $\mathcal{D} \triangleq \mathcal{D}(t)$, $\mathcal{Y} \triangleq \mathcal{Y}(t)$, $x \triangleq x(t)$.

Lemma 3. Assume that $\sum_{k=0}^{\infty} \eta_k = \infty$, $\sum_{k=0}^{\infty} \eta_k^2 < \infty$, and more than one of the following requirements is satisfied

(i) $\mathcal{G}$ is said to be $(2f+1)$-robust with l hops under f-local Byzantine agents

(ii) $\mathcal{G}$ is said to be $(f+1)$-strictly robust with l hops under f-local Byzantine agents, then there exists a row vector q_k^m such that

$$\lim_{k\to\infty} ||\hat{x}_{i,k}^m - \tilde{x}_k^m|| = 0, i \in \mathcal{Q}, \tag{12}$$

$$\lim_{k\to\infty} \eta_k ||\hat{x}_{i,k}^m - \tilde{x}_k^m|| < \infty, i \in \mathcal{Q}, \tag{13}$$

where $\tilde{x}_k^m = q_k^m \hat{X}_k^m$, $\hat{X}_k^m$ denotes the vector consisting of $\hat{x}_{i,k}^m$, and $q_k^m = q_{k+1}^m \bar{L}_k^m$.

Proof: The proof is divided into two parts, considering the conditions (i) and (ii) respectively.

Assume that condition (i) holds, it can be proved by Theorem 2 in this paper and Lemma 3.4 in [23].

If condition (ii) holds, Theorem 3 can imply that the normal agent in graph $\mathcal{G}$ can achieve consensus. Thus, there exists $\tilde{x}_k^m = q_k^m \hat{X}_k^m$ such that (11) and (12) hold.

It can be obtained by (7) that

$$\hat{X}_{k+1}^m = \bar{L}_k^m \hat{X}_k^m + \eta_k \left(I \otimes I_{\{m\}}\right) \Delta_k, \tag{14}$$

in which Δ_k is the vector consisting of $\Delta_{i,k}$. Next, considering that $\bar{L}_k^m$ is a row-random matrix and $\lim_{k\to\infty}\eta_k = 0$ (due to $\sum_{k=0}^{\infty} \eta_k^2 < \infty$), combine (12) and (14), we obtain

$$\lim_{k\to\infty} \bar{L}_k^m \hat{X}_k^m = \lim_{k\to\infty} \bar{L}_k^m \tilde{x}_k^m = \lim_{k\to\infty} 1_M \tilde{x}_k^m.$$

Since $\bar{L}_k^m$ is a row-stochastic matrix, the above equation indicates that q_k^m also is a row-stochastic vector. And

$$\lim_{k\to\infty} \hat{X}_{k+1}^m = \lim_{k\to\infty} 1_{\mathbb{Q}} \tilde{x}_{k+1}^m = \lim_{k\to\infty} 1_{\mathbb{Q}} q_{k+1}^m \bar{L}_k^m \hat{X}_k^m, \tag{15}$$

$$\lim_{k\to\infty} \hat{X}_{k+1}^m = \lim_{k\to\infty} \hat{X}_k^m = \lim_{k\to\infty} 1_{\mathbb{Q}} \tilde{x}_k^m = \lim_{k\to\infty} 1_{\mathbb{Q}} q_k^m \hat{X}_k^m. \tag{16}$$

From (15) and (16), it follows that $\lim_{k\to\infty} 1_{\mathbb{Q}} q_{k+1}^m \bar{L}_k^m \hat{X}_k^m = \lim_{k\to\infty} 1_{\mathbb{Q}} q_k^m \hat{X}_k^m$, which can be derived from $q_k^m = q_{k+1}^m \bar{L}_k^m$. Referring to (11) and $\tilde{x}_k^m = q_k^m \hat{X}_k^m \in [\underline{x}_k^m, \bar{x}_k^m]$, we acquire

$$\begin{aligned} &\eta_{k+r\mathbb{Q}} \left\| \hat{X}_{k+r\mathbb{Q}}^m - \tilde{x}_{k+r\mathbb{Q}}^m \right\| \\ \le &\eta_{k+r\mathbb{Q}} \tilde{a}^r D_k^m + 2\mathbb{Q} \sum_{h=0}^{r-1} \tilde{a}^{r-1-h} \eta_{k+r\mathbb{Q}} \delta_{k+h\mathbb{Q}}. \end{aligned} \tag{17}$$

For the proof of Lemma 8-(b) in [35], (13) can be obtained by (15) and $\sum_{k=0}^{\infty} \eta_k^2 < \infty$. ■

Lemma 3 specifies the graph conditions ensuring that all states $\hat{x}_{i,k}^m$ of normal agents in $\mathcal{Q}$ are consistent. Subsequently, another lemma is introduced to evaluate the reliability of state estimates obtained by normal agents.

Lemma 4. If the requirements in Lemma 3 are satisfied and the vectors $\{q_k^m\}$ in Lemma 3 meet

$$He\left(\bar{O}_k\right) > 0, k \ge 0, \tag{18}$$

where $\bar{O}_k = q_{k+1}\mathcal{I}^T O^T O\left(1_{\mathbb{Q}} \otimes I_n\right)$, then each normal agent generates $\hat{x}_{i,k} = \begin{bmatrix} \hat{x}_{i,k}^1 & \cdots & \hat{x}_{i,k}^n \end{bmatrix}^T \in \mathbb{R}^n$ satisfying $\lim_{k\to\infty} \hat{x}_{i,k} = \lim_{k\to\infty} \tilde{x}_k = x - e_\infty^*$, in which e_∞^* stands for the bounded vector. When $D = 0$, we have $e_\infty^* = 0$.

Proof: Referring to Lemma 3, all the states of normal agents are ensured to achieve consensus as described in (12). Specifically, (12) provides $\lim_{k\to\infty} \eta_k ||\hat{X}_k - \mathcal{I}^T\left(1_{\mathbb{Q}} \otimes I_n\right)\tilde{x}_k|| < \infty$. Because of $\mathcal{I}\bar{L}_k\mathcal{I}^T\left(1_{\mathbb{Q}} \otimes I_n\right) = \left(1_{\mathbb{Q}} \otimes I_n\right)$, it becomes that

$$\sum\nolimits_{k=0}^{\infty} \eta_k ||\mathcal{I}\bar{L}_k \hat{X}_k - \left(1_{\mathbb{Q}} \otimes I_n\right)\tilde{x}_k|| < \infty, \tag{19}$$

where $\tilde{x}_k = q_k \hat{X}_k$, $q_k = q_{k+1}\bar{L}_k$.

In addition, it follows from (14) that

$$\hat{X}_{k+1}^m = \bar{L}_k^m \hat{X}_k^m + \eta_k \left(I \otimes I_{\{m\}}\right) O^T \left(Y - O\mathcal{I}\bar{L}_k \hat{X}_k\right).$$

Then, we have

$$\tilde{x}_{k+1} = \tilde{x}_k + \eta_k q_{k+1}\mathcal{I}^T O^T \left(Y - O\mathcal{I}\bar{L}_k \hat{X}_k\right), \tag{20}$$

where $\hat{X}_{k+1} = \bar{L}_k \hat{X}_k + \eta_k \mathcal{I}^T O^T \left(Y - O\mathcal{I}\bar{L}_k \hat{X}_k\right)$.

Define

$$e_{k+1}^* = \left(I - \eta_k \bar{O}_k\right) e_k^* - \eta_k q_{k+1}\mathcal{I}^T O^T F D, \tag{21}$$

where $e_0^* = 0$.

Based on (18) and $\lim_{k\to\infty}\eta_k = 0$ (since $\sum_{k=0}^{\infty} \eta_k^2 < \infty$), we have $K < \infty$ such that $k > K$,

$$\eta_k \le \eta_K,\ \eta_k\xi_1 - \xi_2 < 0,\ \eta_k^2\xi_1 - \eta_k\xi_2 > -1,$$

with $\xi_1 \ge ||\bar{O}_k||^2$ and $\xi_2 \in \left(0, \lambda_m\left(He(\bar{O}_k)\right)\right)$. Next,

$$||I - \eta_k \bar{O}_k||^2 \le 1 + \eta_k^2 \xi_1^2 - \eta_k \xi_2 \le \left(1 - \eta_k \xi\right)^2, \tag{22}$$

where $0 \le \xi \le \min\left\{\frac{1-\sqrt{1+\eta_K^2\xi_1-2\eta_K\xi_2}}{\eta_K}, \xi_1^{\frac{1}{2}}\right\}$, which ensures that $\eta_k\xi_1 - \xi_2 \le \eta_k - 2\xi$ and $0 < 1 - \eta_k\xi < 1$. Next, from (21) and (22), we have

$$||e_{k+1}^*|| \le ||e_k^*|| + \eta_k\xi\left(Q_M - ||e_k^*||\right).$$

From (21), we obtain $e_{k+1}^* + \eta_k q_{k+1}\mathcal{I}^T O^T \mathcal{Y} = e_k^* + \eta_k \bar{O}_k\left(x - e_k^*\right)$. Let $x_k^* = x - e_k^*$, according to (20), we obtain

$$\begin{aligned}&\left\|\tilde{x}_{k+1} - x_{k+1}^*\right\|^2\\ =&\left\|\tilde{x}_k - x_k^* + \eta_k q_{k+1}\mathcal{I}^T O^T O\left[\left(1_{\mathbb{Q}} \otimes I_n\right)x_k^* - \mathcal{I}\bar{L}_k\hat{X}_k\right]\right\|^2.\end{aligned} \tag{23}$$

Next, set

$$G_1 = \max_{k\ge 0}\left\{\left\|q_{k+1}\mathcal{I}^T O^T O\left[\left(1_{\mathbb{Q}} \otimes I_n\right)x_k^* - \mathcal{I}\bar{L}_k\hat{X}_k\right]\right\|^2\right\},$$
$$G_2 = \max_{k\ge 0}\left\{\left\|2(\tilde{x}_k - x_k^*)^T q_{k+1}\mathcal{I}^T O^T O\right\|\right\},$$

where unknown scalars G_1 and G_2 are bounded (because x, e_k^*, $\hat{X}_k$ and $\tilde{x}_k$ are bounded). By summing both sides of (23),

$$\begin{aligned}&\|\tilde{x}_\infty - x_\infty^*\|^2 + \sum_{k=0}^{\infty}\eta_k(\tilde{x}_k - x_k^*)^T\left(\bar{O}_k + \bar{O}_k^T\right)(\tilde{x}_k - x_k^*)\\ \le& G_1\sum_{k=0}^{\infty}\eta_k^2 + G_2\sum_{k=0}^{\infty}\eta_k\left\|I\bar{L}_k\hat{X}_k - \left(1_{\mathbb{Q}} \otimes I_n\right)\tilde{x}_k\right\|\\ &+ \|\tilde{x}_0 - x_0^*\|^2.\end{aligned} \tag{24}$$

Based on (19), (24) and $\sum_{k=0}^{\infty}\eta_k^2 < \infty$ yield $\sum_{k=0}^{\infty}\eta_k(\tilde{x}_k - x_k^*)^T He\left(\bar{O}_k\right)(\tilde{x}_k - x_k^*) < \infty$.
Because of $\sum_{k=0}^{\infty}\eta_k = \infty$ and (18) provides

$$\eta_k(\tilde{x}_k - x_k^*)^T He\left(\bar{O}_k\right)(\tilde{x}_k - x_k^*) \ge 0,$$

$$\lim_{k\to\infty}\eta_k(\tilde{x}_k - x_k^*)^T He\left(\bar{O}_k\right)(\tilde{x}_k - x_k^*) = 0.$$

At last, we acquire $\lim_{k\to\infty}(\tilde{x}_k - x_k^*) = 0$, which completes the proof by exploiting (12). ■

Remark 11. Refer to [29], Lemma 3 analyzes the consensus of normal agent states and Lemma 4 analyzes the relationship between the state estimate $\hat{x}_{i,k}$ and the system state x. The results show that if the conditions in Lemma 3 and (18) are met, all normal agents will eventually produce an accurate estimate of state x. Lemma 3 and 4 provide theoretical analyses of the MW-DGD algorithm in multi-dimensional states, achieving DSSE of multi-dimensional states while overcoming the dimensional constraints of existing methods under Byzantine agents. However, since the matrix q_k depends on the weight matrix $\bar{L}_{k'}(k' \ge k)$, the condition cannot be directly verified in (18). Therefore, to assess the reliability of the obtained state estimates, it is necessary to evaluate its performance.

VI. Evaluation of Estimation Performance

Section VI introduces a DED algorithm (Algorithm 2) to assess the reliability of state estimates in the presence of Byzantine agents. The DED algorithm is designed to be executed after Algorithm 1.

Algorithm 2 DED algorithm

1: At iteration time k, normal agent i resets $\hat{x}_i = \hat{x}_{i,k}$ and $\beta_{i,0} = ||\mathcal{Y}_i - O_i\hat{x}_i||$, then transmits $h_{ji,0} = (\beta_{i,0}, P_{ji,0})$ to each agent j in $\mathcal{N}_i^{l+}$ at iteration $k = K + cN$. Subsequently, agent i gathers the messages from agents in $\mathcal{N}_i^{l-}$ and itself, forming the set $\mathbb{H}_{i,k} = \left\{h_{ij,k-K-cN} | j \in \mathcal{N}_i^{l-} \cup \{i\}\right\}$.
2: Sort the values in $\mathbb{H}_{i,k}$. Define $\mathbb{W}_{i,k}$ as the largest subset of $\mathbb{H}_{i,k}$ satisfying two conditions: (a) for all $\beta \in B_{i,k} = \mathbb{H}_{i,k}\backslash\mathbb{W}_{i,k}$ and $\beta' \in \mathbb{W}_{i,k}$, $value(\beta) \le value(\beta')$, and (b) the MMC of $\mathbb{W}_{i,k}$ equal f, namely $\left|\mathcal{T}^*\left(\mathbb{W}_{i,k}\right)\right| = f$.
3: Agent i updates the value as
$$\beta_{i,k-K-cN+1} = \max_{\beta \in B_{i,k}}\left\{value\left(\beta_{j,k-K-cN}\right)\right\}.$$
4: Agent i sends $h_{ji,k-K-cN+1}$ to agent $j \in \mathcal{N}_i^{l+} \cup \{i\}$ at iteration $k + 1 < K + (c+1)N$.
5: Return $\{\hat{x}_i, h_{ji,N-1}\}$ at $k = K + (c+1)N - 1$.

The following points are important to note for Algorithm 2.
(i) The value of k in Algorithm 2 is the same as in Algorithm 1. When $k \ge K$ (where K is a sufficiently large integer), Algorithm 2 can operate after Step 4 in Algorithm 1.
(ii) Step 2 in Algorithm 2 in removes the maximum value that makes $\left|\mathcal{T}^*\left(\mathbb{W}_{i,k}\right)\right| = f$, ensuring that Step 3 in Algorithm 2 results in

$$\beta_{i,k-K-cN+1} = \max_{\beta \in \mathcal{Q}}\left\{value\left(\beta_{j,k-K-cN}\right)\right\}. \tag{25}$$

As a result, Byzantine agents cannot expand $\beta_{i,k-K-cN+1}$.
Next, Theorem 4 will be presented to demonstrate that Algorithm 2 can effectively evaluate the estimation performance.
Theorem 4. Consider a multi-agent system $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the set of Byzantine agents $\mathcal{A}$ satisfies Assumption 1, and the normal agents $\mathcal{Q}$ operate according to Algorithms 1 and 2. Let K be a positive integer such that for $k \ge K$,

$$||\hat{x}_{i,k} - \tilde{x}_k|| < \varepsilon, \tag{26}$$

in which ε is the small constant. If $\mathcal{G}$ is $(f+1, s)$-resilient robust with l hops, then $\hat{x}_i (i \in \mathcal{Q})$ in Step 5 in Algorithm 2 satisfies

$$||\hat{x}_i - x|| \le \rho_1(\varepsilon, \beta_{i,N-1}, d_M), \tag{27}$$

$$||\hat{x}_i - x|| \ge \rho_2(\varepsilon, \beta_{i,N-1}, d_M), \tag{28}$$

in which

$$\rho_1 = \max_{|\bar{B}_{i,k}| \ge N-2s}\left\{\varepsilon + b_1\left(|\tilde{B}_{i,k}|\beta_{i,N-1} + c\right)\right\},$$
$$\rho_2 = \max_{j* = \arg\max_{j\in\mathcal{Q}}\{\beta_{j,0}\}}\left\{b_2\left(\beta_{i,N-1} - ||F_{j*}||n^{\frac{1}{2}}d_M\right)\right\} - 2\varepsilon,$$
$b_1 = \lambda_m^{-\frac{1}{2}}\left(\sum_{j\in\tilde{B}_{i,k}} O_j^T O_j\right)$, $b_2 = \lambda_M^{-\frac{1}{2}}\left(O_{j*}^T O_{j*}\right)$,
$c = \sum_{j\in\tilde{B}_{i,k}}\left(||O_j||\varepsilon + ||F_j||n^{\frac{1}{2}}d_M\right)$.

Proof: Let $\Phi_{k-k',k} = L_k \cdots L_{k-k'}(k \ge k')$ and $L_k = [a_{ij,k}]$ with $a_{ij,k} = \sum_{p\in\bar{\mathcal{P}}_{ij,k}} a_{ij,k}^p > 0$ for $i = j$ or $j \in \hat{B}_{i,k}$, in which $\hat{B}_{i,k}$ is the set of agents that produce the value in $B_{i,k}$ and $\bar{\mathcal{P}}_{ij,k} = \{path[h_{ij,k-K-cN}] | h_{ij,k-K-cN} \in B_{i,k}\}$, otherwise, $a_{ij,k} = 0$.

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It is clear that $\beta_{i,k-K-cN+1}$ in Step 3 of Algorithm 2 can be rewritten as

$$\beta_{i,k-K-cN+1} = \max_{\beta \in \bar{B}_{i,k-k',k}} \left\{ value\left(\beta_{j,k-K-cN-k'}\right) \right\}, \quad (29)$$

where $\bar{B}_{i,k-k',k} = \left\{ j \in \mathcal{V} | \left[\Phi_{k-k',k}\right]_i^j > 0 \right\}$. In particular, $\bar{B}_{i,k,k} = \bar{B}_{i,k}$.

Referring to (29) and Step 3 in Algorithm 2, we acquire that

$$\beta_{i,N-1} = \max_{j \in \bar{B}_{i,k-N+2,k}} \left\{ \beta_{j,0} \right\} = \max_{j \in \bar{B}_{i,k-N+2,k}} \left\{ ||\mathcal{Y}_j - O_j \hat{x}_j|| \right\},$$

which implies that

$$\sum_{j \in \tilde{B}_{i,k}} \left\{ ||\mathcal{Y}_j - O_j \hat{x}_j|| \right\} \leq |\tilde{B}_{i,k}| \beta_{i,N-1}, \quad (30)$$

where $\tilde{B}_{i,k} = \mathcal{Q} \cap \bar{B}_{i,k-N+2,k}$. According to (30), we have

$$\begin{aligned} &\sum_{j \in \tilde{B}_{i,k}} \left\| O_j \left(x - \tilde{x}_k \right) \right\| \\ \leq &\left| \tilde{B}_{i,k} \right| \beta_{i,N-1} + \sum_{j \in \tilde{B}_{i,k}} \left(\left\| O_j \left(\hat{x}_i - \tilde{x}_k \right) \right\| + \left\| F_i \mathcal{D} \right\| \right) \\ \leq &\left| \tilde{B}_{i,k} \right| \beta_{i,N-1} + \sum_{j \in \tilde{B}_{i,k}} \left(\left\| O_j \right\| \varepsilon + \left\| F_i \right\| n^{\frac{1}{2}} d_M \right). \end{aligned} \quad (31)$$

Additionally, because the graph $\mathcal{G}$ is called to be $(f+1, s)$-resilient robust with l hops, it can conclude that $|\bar{B}_{i,k-N+2,k}| \geq N - s$ by Lemma 2.

From Assumptions 1 and 2, we get $|\tilde{B}_{i,k}| = |\mathcal{Q} \cap \bar{B}_{i,k-N+2,k}| \geq N - 2s$ and $\sum_{\beta \in \tilde{B}_{i,k}} O_j^T O_j > 0$. Next, based on (31), one acquires

$$||\hat{x}_i - x|| \leq \varepsilon + b_1 \left(|\tilde{B}_{i,k}| \beta_{i,N-1} + c \right), \quad (32)$$

which implies that for $i \in \mathcal{Q}$, $||\hat{x}_i - x|| \leq \rho_1(\varepsilon, \beta_{i,N-1}, d_M)$.

In addition, the following equality can be derived from (25)

$$\beta_{i,N-1} \leq \max_{j \in \mathcal{Q}} \left\{ \beta_{j,0} \right\} = \max_{j \in \mathcal{Q}} \left\{ ||\mathcal{Y}_j - O_j \hat{x}_j|| \right\}, \quad (33)$$

which means that

$$||x - \hat{x}_{j*}|| \geq \frac{\beta_{i,N-1} - ||F_{j*}|| n^{\frac{1}{2}} d_M}{\lambda_M^{\frac{1}{2}} \left(O_{j*}^T O_{j*} \right)},$$

where $j* = \arg\max_{j \in \mathcal{Q}} \left\{ \beta_{j,0} \right\}$. Finally, for $i \in \mathcal{Q}$,

$$\begin{aligned} ||\hat{x}_i - x|| \geq &||\hat{x}_{j*} - x|| - ||\hat{x}_{j*} - \tilde{x}_k|| - ||\tilde{x}_k - \hat{x}_i|| \\ \geq &b_2 \left(\beta_{i,N-1} - ||F_{j*}|| n^{\frac{1}{2}} d_M \right) - 2\varepsilon, \end{aligned} \quad (34)$$

which implies that for $i \in \mathcal{U}$, $||\hat{x}_i - x|| \geq \rho_2(\varepsilon, \beta_{i,N-1}, d_M)$. ■

Theorem 4 demonstrates that Algorithm 2 provides a valid approach for evaluating the estimation performance of normal agents. First, (27) indicates that when ε is sufficiently small (when K is large enough) and $\beta_{i,N-1}$ is small, the state estimate $\hat{x}_i$ obtained by agent i is reliable. Furthermore, (28) shows that if all normal agents provide reliable state estimates, the obtained $\beta_{i,N-1}(i \in \mathcal{Q})$ is always not big. In other words, if K is sufficiently large and $\beta_{i,N-1}$ is too big, agent i identifies that the agents are Byzantine, thereby reducing the estimation performance.

Remark 12. Our preliminary directions are threefold: (i) Replace each iteration updates with iteration intervals and introduce the definition of jointly r-strictly robust with l hops on the union graph of each interval. (ii) Adapting the sorting and filtering methods, along with the MMC concept, to operate over these iteration intervals. The weights would then be defined based on the number of valid paths within a interval. (iii) Establish mathematical equivalent representation for the time-varying case and then derive convergence error bounds under jointly r-strict robustness with l hops. The detailed algorithm and its rigorous proof will be systematically developed in future work.

VII. Demonstrative Example

This section draws on [29] to consider an unmanned ground vehicle (UGV) system. By assuming that the vehicle moves in a straight line and only turns when it stops moving in a straight line, its system can be described as follows

$$\begin{aligned} &\begin{bmatrix} \dot{s} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = A \begin{bmatrix} s \\ v \\ \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{M} & 0 \\ 0 & 0 \\ 0 & \frac{1}{J} \end{bmatrix} \begin{bmatrix} F \\ T \end{bmatrix} + d(t), \\ &y_i(t) = C_i x(t), i \in \mathcal{Q}, \end{aligned} \quad (35)$$

where $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{b}{M} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{b_r}{J} \end{bmatrix}$, $d_1(t) = \begin{bmatrix} \cos(\pi t) \\ \cos(\pi t) \\ \cos(\pi t) \\ \cos(\pi t) \end{bmatrix}$, $d(t) = 0.01 d_1(t)$ is the noise satisfying $\|d(t)\| \leq 0.04$, $x = \begin{bmatrix} s & v & \theta & w \end{bmatrix}^T$ represents the system state (s, v, θ, and w describe position, linear velocity, angular position, and angular velocity, respectively), torque T and force F denote system inputs. $J = 2$, $M = 1$, $b = 1$ and $b_r = 1$ represent mass, inertia, translational friction, and rotational friction coefficient, respectively. Suppose that our initial value is $x(0) = \begin{bmatrix} 5 & 0_{1\times 3} \end{bmatrix}^T$, and then the UGV follows the following trajectory: (i) shifts to the point in $x = 5$ or $x = -5$, (ii) stops the straight line motion after reaching it, turns 180°, (iii) and repeats the above motion.

As shown in Fig. 3, consider four multi-agent systems below: (i) Consider $N = 7$ agents, and $C_i = \begin{bmatrix} i \times I_2 & 0 \end{bmatrix}$, $C_{i+3} = \begin{bmatrix} 0 & i \times I_2 \end{bmatrix}$, $C_7 = \begin{bmatrix} 0 & I_2 \end{bmatrix}$, $i \in \{1, 2, 3\}$. (ii) Consider $N = 17$ agents, and $C_{i+3\times j} = \begin{bmatrix} i \times I_2 & 0 \end{bmatrix}$, $C_{i+3\times z} = \begin{bmatrix} 0 & i \times I_2 \end{bmatrix}$, $i \in \{1, 2, 3\}$, $j \in \{0, 1, 2\}$, $z \in \{3, 4\}$, $C_{16} = \begin{bmatrix} 0 & I_2 \end{bmatrix}$, $C_{17} = \begin{bmatrix} 0 & i \times I_2 \end{bmatrix}$. (iii) Consider $N = 6$ agents, and $C_i = \begin{bmatrix} i \times I_2 & 0 \end{bmatrix}$, $C_{i+3} = \begin{bmatrix} 0 & i \times I_2 \end{bmatrix}$, $i \in \{1, 2, 3\}$. It can be seen that each agent cannot estimate the global state independently, so it is necessary to adopt a DSSE method.

A. Graph (a): 7-agent Network

The network in Fig. 3(a) is 2-strictly robust under two hops rather than 2-strictly robust under one hop. Assume agent 7 represents a Byzantine agent (if an agent is Byzantine, its update strategy (6) should include an additional value $4\sin(k) -$

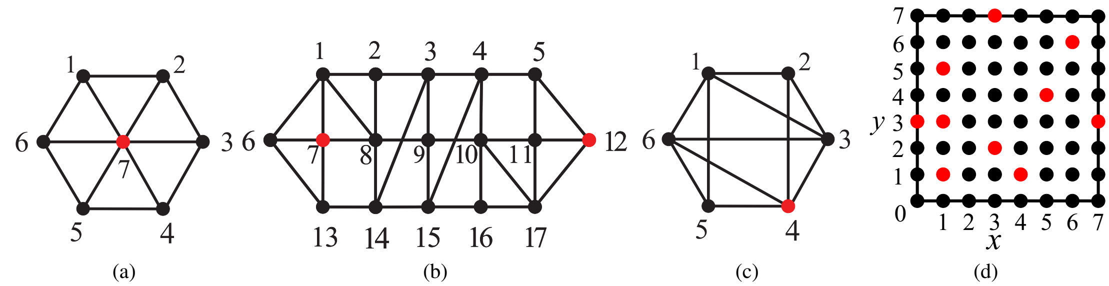

Fig. 3. The communication graphs (If two agents are connected, they can collect messages from each other).

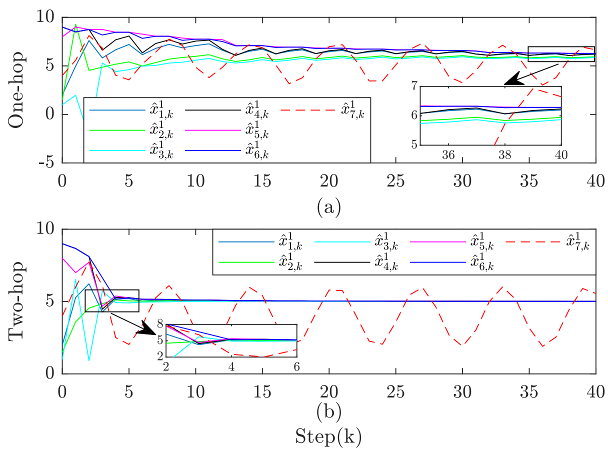

Fig. 4. The estimates of $x(t)$ for Fig. 3(a) with 7 agents (Byzantine agent $\mathcal{A} = \{7\}$).

1). Agent 7 can transmit distinct values to each of its six neighbors. Set the initial state as $\hat{x}_k^m = [2,\ 1.5,\ 1,\ 9,\ 8,\ 9,\ 4]^T$, with $m = \{1, \cdots, 4\}$. Let $\eta_k = 0.15/k$ and $K = 30$. Suppose agent 7 adds all values sent to agents 1, 2, 3, and 4 (including its own values and the relayed values) to an additional $5 + 9\sin(2k)$, and all values sent to agents 5, 6, and 7 (including its own values and the relayed values) to an additional $5 - 9\cos(2k)$.

1) Inefficiency analysis of consensus and convergence under one-hop communication: Since the graph is not 2-strictly robust with one hop, consensus may not be achievable among normal agents using one-hop communication. This can be verified by the state values shown in Fig. 4(a). According to Algorithm 2, when $k \geq K$, normal agents exhibit $\beta > 10$, suggesting that the state estimates of normal agents in $\mathcal{Q}$ are unreliable. Fig. 4(a) illustrates that $\hat{x}_{i,k}^1$ does not converge to the state $x^1(0) = 5$, confirming that Algorithm 2 effectively evaluates the reliability of the obtained state estimates.

2) Efficiency analysis of consensus and convergence under two-hop communication: We investigate the two-hop weighted distributed gradient descent algorithm. Based on the state values shown in Fig. 4(b), we observe that consensus is achieved through two-hop communication. Specifically, at step $k = 10$, agent 1 receives 9 values and removes 3 of them using Algorithm 1. Among these removed values, 2 values from Byzantine agent 7 or are corrupted by it during transmission, while the remaining 1 values are normal. At step $k = 10$, Algorithm 1 successfully removed all adversarial values, and then the agents to reach consensus. According to Algorithm 2, when $k \geq K$, normal agents in $\mathcal{Q}$ satisfy $\beta \leq 1.2$, indicating that their state estimates are reliable. For instance, Fig. 4(b) illustrates that $\hat{x}_{i,k}^1$ converges to a consensus value of 5.0037.

Remark 13. The network in Fig. 3(a) is not 2-strictly robust with 1 hop, but is 2-strictly robust with two hops. To illustrate, suppose agent 7 is removed. Consider the two disjoint agent sets $\mathcal{V}_1 = \{1, 2, 3\}$ and $\mathcal{V}_2 = \{4, 5, 6\}$. Under one-hop communication, none of the agents in $\mathcal{V}_1$ has two or more neighbors outside $\mathcal{V}_1$. The same holds for set $\mathcal{V}_2$. Therefore, the remaining subgraph is not 2-robust with 1 hop. According to Definition 5, the 7-node network is therefore not 2-strictly robust with 1 hop. However, the conclusion changes when considering two-hop communication. Again, removing agent 7 and considering sets $\mathcal{V}_1$ and $\mathcal{V}_2$, agent 2 now has two independent paths originating from nodes outside $\mathcal{V}_1$ (namely, the paths $6 \to 1 \to 2$ and $4 \to 3 \to 2$). After checking all possible set partitions, we confirm that the remaining subgraph meets the condition for being 2-robust under $l = 2$. Furthermore, we can demonstrate that the network remains 2-strictly robust with 2 hops when any single node is removed. Thus, the 7-agent network is 2-strictly robust with 2 hops.

B. Graph (b): 17-agent Network

Consider the network of 17 agents shown in Fig. 3(b). It is not 2-strictly robust for one-hop communication but is 2-strictly robust for two-hop communication. Assume agents 7 and 12 are Byzantine agents whose update policy (6) includes an additional value $-0.5 + rand(4, 1)$. Furthermore, assume that agents 7 and 12 add all values sent to agents 1-8 (including their own values and the relayed values) to an additional value $5 + 9\sin(2k)$, and all values sent to agents 9-17 (including their own values and the relayed values) to an additional value $5 - 9\cos(2k)$. Let the initial normal state be $\hat{x}_k^m = [2,\ 1.5,\ 1,\ 3,\ 2,\ 1,\ 2.5,\ 4,\ 8,\ 6,\ 7,\ 7,\ 9,\ 8,\ 9,\ 6,\ 7]^T, m = \{1, \cdots, 4\}$. Set $\eta_k = 0.15/k$.

1) Inefficiency analysis of consensus and convergence under one-hop communication: Since the graph is not 2-strictly robust with one hop, Fig. 5(a) shows that consensus cannot be achieved for one-hop communication. Referring to Algorithm 2, when $k \geq K$, normal agents display $\beta > 8$, indicating that the state estimates of the normal agents in $\mathcal{Q}$ cannot reach convergent. Fig. 5(a) shows that $\hat{x}_{i,k}^1$ does

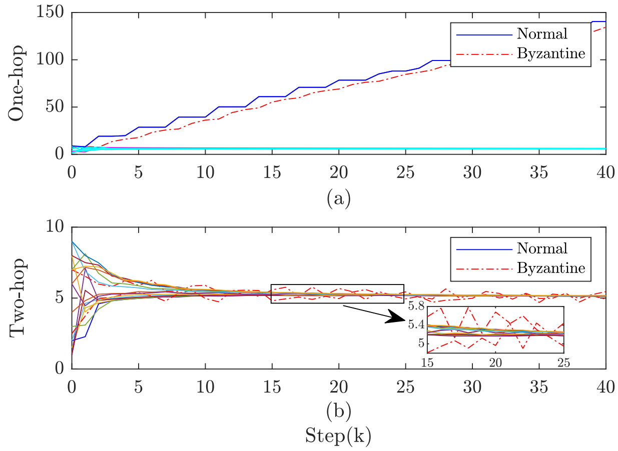

Fig. 5. The estimates of $x(t)$ for Fig. 3(b) with 17 agents (Byzantine agents $\mathcal{A} = \{7, 12\}$).

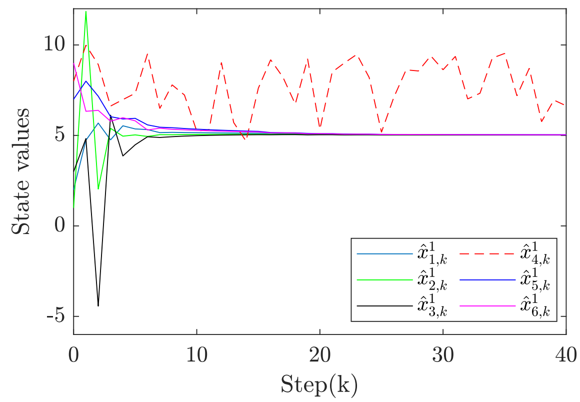

Fig. 6. The estimates of $x(t)$ for Fig. 3(c) with 6 agents (Byzantine agent $\mathcal{A} = \{4\}$).

not converge to $x^1(0) = 5$, validating the effectiveness of Algorithm 2.

2) **Efficiency analysis of consensus and convergence under two-hop communication:** The two-hop weighted distributed gradient descent algorithm proposed is used for simulation. As given in Fig. 5(b), consensus is achieved. Based on Algorithm 2, when $k \geq K$, normal agents in $\mathcal{Q}$ have $\beta \leq 2$, suggesting that their state estimates are reliable. For example, Fig. 5(b) shows that $\hat{x}^1_{i,k}$ converges to a consensus value of 5.0247.

C. Graph (c): 6-agent Network

Assuming that agent 4 is a Byzantine agent, an additional value $4\sin(k) - 1$ must be incorporated into its update policy (6). Furthermore, assume that agent 4 introduces an extra value $5 + 9\sin(2k)$ to all values sent to agents 1-3 (including their own values and forwarded values), and introduces an extra value $5 - 9\cos(2k)$ to all values sent to agents 4-6 (including their own values and forwarded values). Let $\hat{x}^m_k = [2,\ 1,\ 3,\ 8,\ 7,\ 9]^T, m = \{1, \cdots, 4\}$ and $\eta_k = 0.25/k$.

1) **Validity analysis of Theorem 3:** For the fixed hop number $l = 1$, Fig. 3(c) is 2-strictly robust but not 3-robust under 1-local Byzantine model. This graph requires additional edges to become a 3-robust graph. Fig. 6 demonstrates that the agents achieve consensus, verifying the correctness of the necessary and sufficient conditions in Theorem 3. Moreover, as Fig. 3(c) achieves consensus under the $(f + 1)$-strictly robust condition but not $(2f+1)$-robust condition, it indicates that Theorem 3 imposes less restrictive graph connectivity requirements than Theorem 2.

2) **Different validation aspects of Graph (a)-(c):** Based on the above discussion, the 7-agent and 17-agent networks demonstrate that the multi-hop algorithm outperforms the one-hop algorithm. The simulation results indicate that these networks fail to achieve consensus under one-hop communication but succeed under two-hop communication. This is because the networks in Fig. 3(a) and Fig. 3(b) are 2-strictly robust with 2 hops but not with 1 hop. Furthermore, the 6-agent network confirms the correctness of the necessary and sufficient conditions in Theorem 3 and illustrates that Theorem 3 relaxes the graph condition constraints of Theorem 2. Specifically, for a fixed hop number l, Fig. 3(c) achieves consensus under the $(f + 1)$-strictly robust condition but not under the $(2f + 1)$-robust condition.

D. Graph (d): 64-agent Network

A wireless sensor network comprising 64 agents is constructed, where the agents are uniformly deployed in an 8×8 grid, as shown in Fig. 3(d). Agents are indexed by $\{0, 1, \ldots, 63\}$, and the coordinates of agent i are $\big(i \bmod 8,\ \lfloor i/8 \rfloor\big)$. We set a set of 10 Byzantine agents with indices $[9, 19, 24, 25, 59, 12, 31, 37, 41, 54]$, highlighted in red in Fig. 3(d). In the sensitivity experiments, the value of f is varied from 0 to 10. For a given radius r, the network topology is formed by connecting any two agents within Euclidean distance r with a bidirectional communication link. For each (f, r) pair, the topology is fixed and MW-DGD algorithm is run with $l \in \{1, 2, 3\}$ to compare performance across f, r, and l.

The initial values of normal agents are randomly selected from the interval [0, 20]. Byzantine behavior is defined as follows (with k the iteration index): agents with odd indices add either $\cos k$ or $\sin k$ to all transmitted values (including its own state and relayed values), whereas agents with even indices add either $\sin k$ or the constant 0.5. For each (f, r), we conduct 20 independent trials. After $K = 100$ iterations, we compute the consensus error. A trial is considered successful if this error is less than 0.5; otherwise, it is deemed a failure. Consequently, this criterion classifies cases with excessively slow convergence as failures. The success rates of the algorithm under different (f, r) combinations for one-hop, two-hop, and three-hop settings are plotted in Fig. 7.

We analyze the results from three perspectives: the number of Byzantine agents f, the number of hops l, and the communication radius r. (i) For fixed l and r, the success rate of achieving consensus decreases as f increases.(ii) For any given f and r, the success rate increases with the number of hops l. This improvement is particularly significant when

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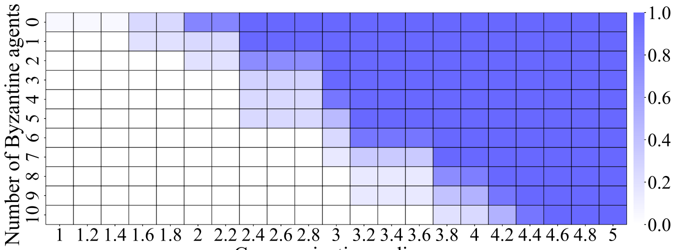
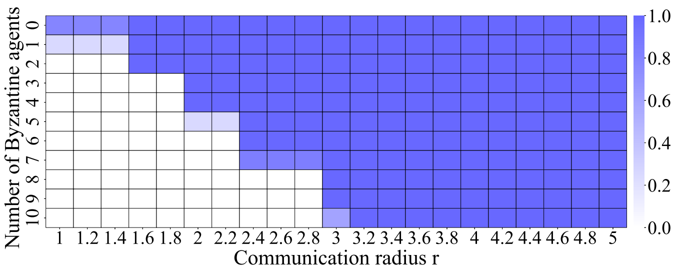
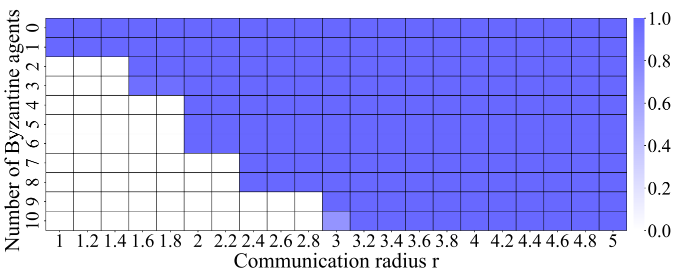


(a) One-hop communication



(b) Two-hop communication



(c) Three-hop communication

Fig. 7. Resilient consensus success rate.

$f \leq 7$ and $r \leq 2.8$, which validates that the graph robustness increases with l. (iii) For any fixed f and l, increasing r raises the success rate by enhancing network connectivity and path redundancy. Based on this analysis, consensus can be accelerated by increasing the number of hops and the communication radius. Therefore, incorporating multi-hop communication in the MW-DGD algorithm not only enhances network robustness but also accelerates the convergence of consensus, even in adversarial environments.

VIII. Conclusion

This paper has investigated the problem of DSSE for CPSs prone to Byzantine attacks. To address this challenge, a novel MW-DGD algorithm has been proposed, which achieved secure estimation in polynomial time. Theoretically, a necessary and sufficient condition for achieving consensus under the f-local Byzantine model has been established. Meanwhile, the convergence analysis has demonstrated that the estimated values of normal agents converged to the true system state. Additionally, a more universal distributed error detection algorithm, incorporating the minimum message coverage concept, has been introduced for dynamic performance evaluation in broader multi-hop communication. At last, simulation examples have validated the effectiveness of our method. Note that time-varying networks allow communication links to change over time and better reflect real-world complexity. Consequently, extending the proposed method to such settings is a valuable research direction.

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